CARDIOLINE

touchECG

User manual

CE 1936

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CARDIOLINE

Cardioline Spa
Via Linz, 151
38121 Trento
Italy

Contents

1. GENERAL INFORMATION

This manual is an integral part of the device and should always be available as support material to the clinical practitioner or the operator. Strict compliance with the information contained in this manual is an essential prerequisite for the proper and reliable use of the device.
Have the operator read the manual thoroughly, as a great deal of the information is only described once.

1.1. Minimum Requirements for the computer/tablet/smartphone

touchECG can be installed on any computer (PC, tablet, notebook, etc.) with the following minimum requisites:

Operating System	Windows:	Windows 10
	Android:	Android 4.4 KitKat (API 19) or higher
Processor	Quad core 1.6 GHz or higher	
RAM	Windows:	2GB or more
	Android:	1 GB or more
Free space on Hard Disk	8 GB or more	
Monitor	Windows:	640 x 480 pixels or more
	Android:	Tablet: 7” or more Smartphone: Samsung 4.7” or more
Bluetooth	Bluetooth 2.1 +EDR for HD+; Bluetooth >= 4.2 for HD+12/HD+ 15 at 500 Hz Bluetooth >= 4.2 for HD+12/15 at 500 Hz Bluetooth >= 4.2 with 2MPhy for HD+12/HD+ 15 at 1000 Hz	
Printer	Laser (colour/BW)	
Additional applications	Windows:	Email application which supports the EML format (only required for the email File Upload feature)
	Android:	Acrobat “PDF” file format viewer

1.2. Licensing terms

By installing the software, the terms and conditions described as follows are accepted.
Object of this agreement is the consent of a use licence for the software and the operating manual. Cardioline SpA guarantees a personal licence, non-exclusive and non-transferable, for use of the software and the attached documents. The software and accompanying documents are protected by copyright. The user must comply with copyright law dispositions.
All rights relative to the software are the property of Cardioline SpA. The transferral of software onto another computer by means of the network or data channel is not permitted.

The programs and the attached documents cannot be changed, copied, merged with other programs or made available to third parties.
The user is deemed responsible for any damage stemming from non-compliance with the copyright, or from violation of the conditions reported in this agreement.

1.3. Other important information

This manual was written with the utmost care. Should you find any details which do not correspond to those contained in this manual, please inform Cardioline SpA, who will proceed to correct such inconsistencies as soon as possible.

The information contained in this manual is subject to change without notice.
All changes will be in compliance with the regulations governing the manufacturing of medical equipment.

All trademarks mentioned in this document are property of their respective owners. Their protection is guaranteed.
No part of this manual may be reprinted, translated or reproduced without the manufacturer's written authorisation.

The codes relating to this manual are listed below.

Language	Code
English	36519163_EN

2. SAFETY INFORMATION

Cardioline SpA will be held responsible for the safety, reliability and functionality of the devices only if:

1. Assembly, modification or repair operations are carried out by Cardioline SpA or by one of its Authorised Assistance Centres;
2. The device must be used in compliance with the instructions contained in the user manual.

Always contact Cardioline SpA should you wish to connect any devices not mentioned in this manual.

Warnings

- This manual provides important information on the proper use and safety of the device. Failure to comply with the described operating procedure, improper use of the device, ignoring the specifications and recommendations supplied, may cause severe physical injuries to the operators, patients and bystanders, or may damage the device.
- The device cannot be modified in any way.
- Do not use the device if you suspect it is malfunctioning.
- The device captures and presents the data that reflects the physiological condition of the patient; this information can be examined by specialist medical staff and will be useful in providing an accurate diagnosis. In any event, the data cannot be used as the only means to make an accurate diagnosis of the patient.
- The operators for whom this device is intended must have the required competence regarding medical procedures and the treatment of patients. They must also be sufficiently trained in using the device. Have the operator carefully read and understand the contents of the operator manual and the other annexed documents before using the device for clinical applications. Inadequate knowledge or training could be at a greater risk for the physical safety of operators, patients and bystanders, or could damage the device.
- US Federal Legislation restricts the sale of this device to prescription by a doctor.
- The device is designed for use only with the HD+ acquisition unit. Refer to the HD+ user manual for the risks and warnings associated with it and for its user instructions.
- This device may be installed on a tablet, desktop computer or laptop (hereinafter the support device), which satisfies the minimum requirements indicated in para. 1.1. To ensure the electrical safety of the operator and patient during its operation, the following precautions must be observed:
    - If the support device is battery powered, do not connect it to an external power supply (for recharging) or other electrical equipment (e.g. to another computer via USB or to a LAN network) when using it in the patient area.

  - If the support device is powered off the mains, it may not be used in the patient area. When used in the patient area, it must be powered with an isolation transformer and shielded cables must be used to connect it to the LAN. The power cable shielding (when present) must be connected to an earthing system appropriate for the area where the device is used. This will avoid electric shocks caused by different earth potentials which could exist between the various points of an electricity distribution system, or else by failures of the external equipment connected to the mains.

- The HD+ acquisition unit for use with the device is protected against defibrillation. Check the HD+ unit's cables for cracks and breakage before using it.
- Refer to the HD+ unit's user manual for the risks and warnings associated with it.
- This device is designed to be used only with the electrodes specified in this manual. Strictly follow the correct clinical procedures to prepare the skin before the application of the electrodes and monitor the patient in order to avoid any irritation, inflammation or other skin reactions. The electrodes are designed for short-term applications and must be promptly removed once the examination is complete.
- The ECG electrodes may cause skin irritation; check the skin for any irritations or inflammations.
- To prevent any infections, use the disposable components (e.g. the electrodes) only once. To ensure safety and use efficiency, do not use electrodes after their expiration date.
- The quality of the signal produced by the device may be adversely affected by the use of other medical equipment such as defibrillators and ultrasound machines.
- The risks associated with the use of the device together with other equipment, such as pacemakers and other stimulators, depend on the support device on which it is installed. The signal may be subject to disturbance in any case.
- The use of the device in combination with other high frequency (HF) surgical equipment depends on the support device in question.
- The operation may be adversely affected by the presence of strong magnetic fields such as those produced by electro surgery equipment.
- The use of the device in the presence of diagnostic imaging equipment, such as Magnetic Resonance (MR) or Computer Axial Tomography (CAT), in the same room, depends on the support device in question.
- The software reports the battery level of the HD+ unit. The low battery warning is designed to work solely with the HD+ unit using the batteries indicated in its user manual. If the battery level is low, replace the HD+ unit's batteries and observe its user instructions.
- Using a support device equipped with a GPRS or WLAN unit may interfere with other nearby equipment. Check with local authorities or with providers of the spectrum of your structure to find out whether any restrictions apply to use of the device in your area.
- Do not leave the patient cable unattended in the presence of children as they could be accidentally strangled.
- Do not leave the electrodes unattended in the presence of children as they could cause suffocation if accidentally swallowed.

- If you are printing PDF files, you need to set up a corresponding program so that the document is in no way adapted or scaled. If you are using Acrobat Reader, you need to select "Actual Size" in "Page management and size." Otherwise you may get nondiagnostic quality prints.
- In the case of Android, we recommend using printers approved by Cardioline, for which the diagnostic quality of prints is ensured.

Attention

- The device does not require any calibration or special instrumentation for correct use and maintenance.
- The Glasgow algorithm is intended for automatic interpretation of ECGs at rest. The automatic interpretation provided by touchECG can only be considered if HD+ is used to acquire ECGs at rest. If HD+ is used to acquire signals while the patient is moving (for example during a stress test) the automatic interpretation may not be valid.
- The Glasgow 12-lead resting ECG Analysis Algorithm has specific criteria for patients of different ages, gender and race. If the option is enabled, the algorithm can provide the doctor with an automatic interpretation, which generates diagnostic messages on the ECG report. Validation is required by a cardiologist or a doctor.

Notes

- The movements of the patient may generate excessive noise and affect the quality of the ECG tracing or the correct analysis of the device.
- An appropriate preparation of the patient is important in order to guarantee a proper application of the ECG electrodes and the correct operation of the device.
- The incorrect positioning of the electrodes for the detection of the algorithm depends on the normal physiology and on the order of the ECG leads and tries to identify the most likely exchange. However, it is recommended to check the positioning of the electrodes of the same group (limbs or chest).
- If the electrodes are not properly connected to the patient, or one or more patient leads are damaged, the display will indicate that the leads in question are disconnected. When the ECG is printed, these leads will come out on paper as a square wave.
- The accuracy of the measurements taken by this device complies with the specific requirement of standard IEC 60601-2-25.
- The device is a Class IIa in compliance with Directive 93/42/EEC.
- The device is a "prescription device" under the terms of FDA regulations.
- The HD+ unit must be connected to the support device running the software before use.

2.1. Warnings on Bluetooth

2.1.1. Potential sources of interference of Bluetooth

It is important to learn how to minimise wireless interference that may cause slow performance and disconnections from Bluetooth devices.
Specifically, there might be interferences and it is required to make an operation check if the following are found:

- Inability to couple a Bluetooth device to the computer/tablet
- Intermittent HD+ disconnections
- Loss of the ECG signal.

Possible causes of interference of the Bluetooth signal may be:

- Microwave ovens: using microwave ovens close to the computer/tablet may cause interference.
- Satellite TV: the coaxial cable and connectors used with some satellite dishes may cause interference. Check whether the cables are damaged thus causing radio interferences (RF leakage). Replace the cables if interference is suspected.
- Power supply: some external electricity sources such as power lines, electrical railroad tracks and power stations, may cause interference. Avoid placing the computer/tablet near wall power lines or close to junction boxes.
- 2.4 GHz or 5 GHz telephones: cordless telephones operating in the 2.4 GHz or 5 GHz range may cause interference with other wireless devices when used to place calls.
- Wireless RF video: wireless video transmitters that operate in the 2.4 GHz or 5 GHz range may cause interference with other wireless devices.
- Wireless speakers: wireless speakers that operate in the 2.4 GHz or 5 GHz range may cause interference with other wireless devices.
- Certain external VDUs and LCD displays: certain displays emit harmonic interferences, especially in the 2.4 GHz band between channels 11 and 14. These interferences may be stronger if using a notebook computer with a closed display and a connected external monitor.
- Inadequately shielded cables: external hard disks or other devices with poorly shielded cables can interfere with wireless devices. If disconnecting or switching off the device reduces the interferences, try replacing the cable that connects the device to the computer/tablet.
- Other wireless devices: other wireless devices that operate in the 2.4 GHz or 5 GHz band (nearby microwave transmitters, wireless cameras, baby monitors, WiFi devices) may cause interferences with Bluetooth connections.
- **Enabling GPS operation**: if the GPS function is enabled on the device in use, it may create interference with the Bluetooth signal. If losses of Bluetooth signal packets are noted, it is recommended to disable the GPS function.

2.1.2. Obstacles for the wireless signal

The position of the device and the construction materials may affect Bluetooth performance. If possible, avoid obstacles or move the Bluetooth device for the signal to travel through fewer obstacles. For example, avoid placing the computer/tablet under a metal desk and the HD+ device on top of a desk.

Below is a chart showing the radio signal-blocking ability of the most common devices:

Material	Potential interference
Wood	Low
Synthetic material	Low
Glass	Low
Water	Medium
Bricks	Medium
Marble	Medium
Plaster	High
Concrete	High
Bulletproof glass	High
Metal	Very high

2.1.3. Reducing the effects of the interference of other wireless devices

If a lot of wireless devices are connected to the computer/tablet or nearby, it may be required to encode the channels used by the WiFi devices.
To reduce the interference between WiFi devices and Bluetooth to the minimum try these solutions:

- Change channels of the wireless network;
- Connect to a 5 GHz wireless network (if possible);
- Place the HD+ close to the computer/tablet.
- Reduce to the minimum the number of Bluetooth devices connected to the computer/tablet or nearby.

3. SYMBOLS AND LABEL

3.1. Explanation of the symbols

Symbol	Description
	Comply with the instructions in the use manual
CE 1936	CE marking – compliance with the European Union directives
	Refer to the user instructions
	Manufacturer

3.2. Device label

Windows

Android

4. INTRODUCTION

4.1. Purpose of the manual

This manual covers the touchECG product.
The manual represents a guide for the execution of the following operations:

- Reasonable use of the device, of the function keys and of the sequence of menus.
- Preparation of the device for use (Section 5)
- Acquisition, printing and storage of ECG traces (Section 6)
- Connectivity and transmission of ECG tracings. (Section 7)
- System configuration (Section 8)
- Updating the device (Section 10)
- Troubleshooting (Section 11)

4.2. Recipients

This manual is intended for professional healthcare operators. They are therefore presumed to have specific knowledge of medical procedures and terminology, as required by clinical practice.

4.3. Intended use

touchECG is designed to monitor and diagnose cardiac function. However, a Cardiologist must validate the results of the analysis run by the ECG.
touchECG is intended for use in hospitals, clinics and outpatient departments of any size. It is also suitable for use by medical staff and trained operators who operate on behalf of a doctor at the patient's home and in an emergency (ambulance).

touchECG is intended for use with the devices of the series Cardioline HD+ (HD+, CLICKECG-HD, HD+12, HD+15, CLICKECG-HD 12, CLICKECG-HD 15). The Cardioline HD+ device acquires the ECG signal and sends it via Bluetooth to the PC on which the touchECG software is installed. touchECG software is designed to only work with the Cardioline HD+ device.

- The device acquires, analyses, displays and prints out electrocardiograms.
- The device interprets the data for review by a doctor.
- The device is intended for use by a doctor or by healthcare personnel on behalf of an authorised doctor in clinical facilities. It is not intended as the only means for determining the diagnosis.
- The device's interpretation of the ECG analysis is only significant if used together with an additional analysis by a physician and with an assessment of all the patient's other significant data.

- The device can be used on adult and paediatric patients.
- The device must not be used as a physiological monitoring of vital signs.

4.4. Description of the device

touchECG is a software that implements a 12- or 15-lead diagnostic electrocardiograph. The ECG signal is acquired through the acquisition unit of the series HD+ (HD+, HD+ 15, HD+ 12, CLICKECG-HD 12, CLICKECG-HD 15) and is transmitted thereby, via Bluetooth or via USB, to the computer on which the touchECG software is installed. touchECG, therefore, makes it possible to acquire, view, print, store and review the ECG tracings received, for adults and children, calculating the principal global ECG parameters.
Depending on the acquisition unit used, touchECG can receive and manage 12- and 15-lead ECG tests:

- HD+, HD+ 12, CLICKECG-HD 12: 12 leads;
- HD+ 15, CLICKECG-HD 15: 15 leads.

The device can be supplied with the optional 12- or 15-lead Glasgow resting ECG interpretation algorithm, with specific criteria for patients of different age, sex and race. If this option is enabled, the algorithm can provide the referring physician with an additional automatic report, generating diagnostic messages in the ECG report. This automatic interpretation must always and however be interpreted by the doctor and may not be used as sole diagnosis.
For further information on the resting ECG interpretation algorithm, see the Instruction Manual for doctors for its use with adults and children which is supplied with the device.

The device can be configured with the DICOM® function.

The device installs on any PC, tablet, smartphone, notebook with the minimum requisites listed in para. 1.1.
It prints out in the following formats: standard or Cabrera 3, 3+1, 3+3, 6, 12 or 15 channels in automatic mode, and 3, 6, 12 or 15 printout channels of the rhythm strip.

The touchECG device includes:

1. USB stick with touchECG software and operating manual
2. Information brochure

The device can be sold as a system that may include, depending on the sale configurations:

1. HD+ unit
2. Computer (tablet or all-in-one PC)
3. Printer
4. Trolley (tablet model or digital model)
5. Power bank.

Each device in the system comes with its own user manual and accessories.

***Note**: the Glasgow Algorithm Interpretation is always loaded on the device but can be enabled or disabled according to the acquired options. The Glasgow interpretation program can only be enabled or disabled by using the activation code provided by Cardioline (see Para. 10.2).*
The Glasgow interpretation program runs the automatic interpretation as well as the readings on the ECG traces. In any case, if the Glasgow interpretation program is disabled, only the readings are printed, whereas the automatic interpretation is not printed.

4.5. General overview

The program windows are divided into three main areas:

- Top bar (1):

  at the top, it displays the window's main information, e.g. patient first and last name, messages and so on.

- Side bar (2):

  on the right, it contains the window's function buttons.
  Two different displays are available, that may be selected from the Settings page (see para. 8.2.6):

  - "Complete user interface", showing all available controls,
  - "Quick user interface", showing some controls only.

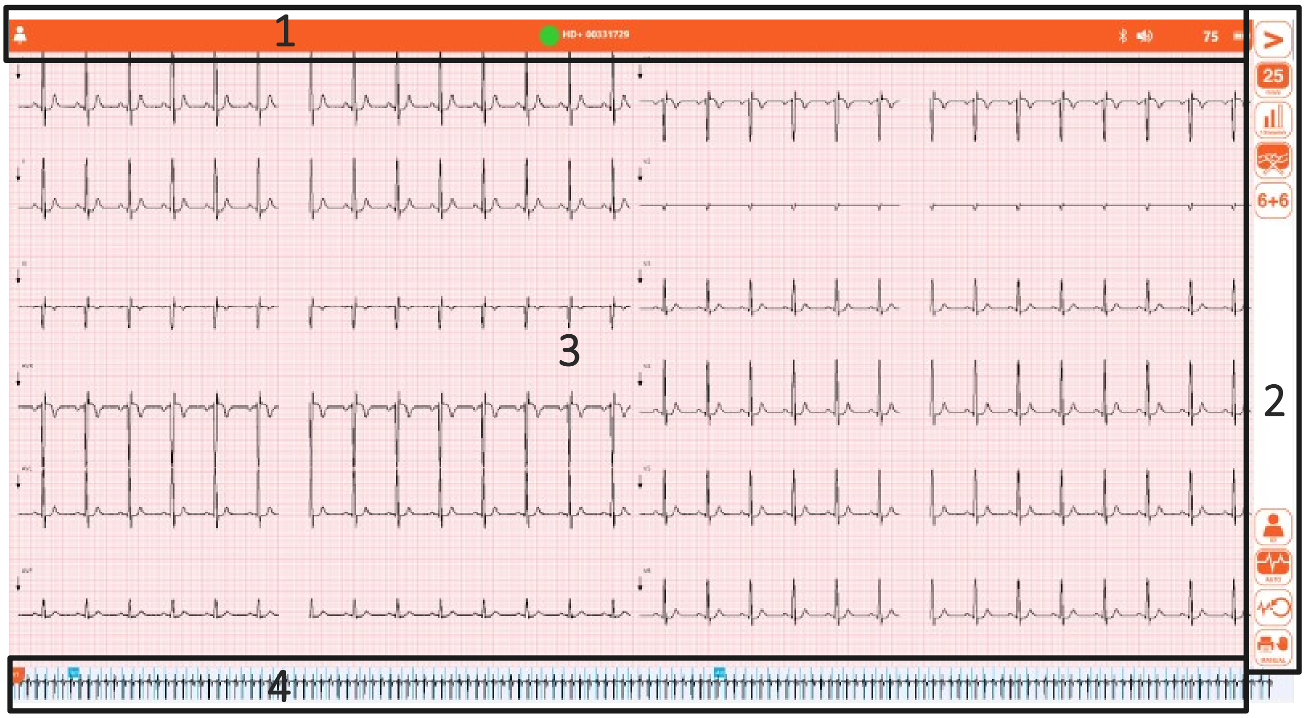
  Some buttons, such as trace speed, width, filter and so on, allow you to select parameter values and change when clicked on to reflect the current value. The trace speed button, for instance, allows you to select a different value with each click, and cycles through the possible rates (5, 10, 25, 50 mm/s). It changes to display the currently selected value (5, 10, 25 or 50).
  Other buttons activate/deactivate their value and change to reflect this.

- Central area (3):

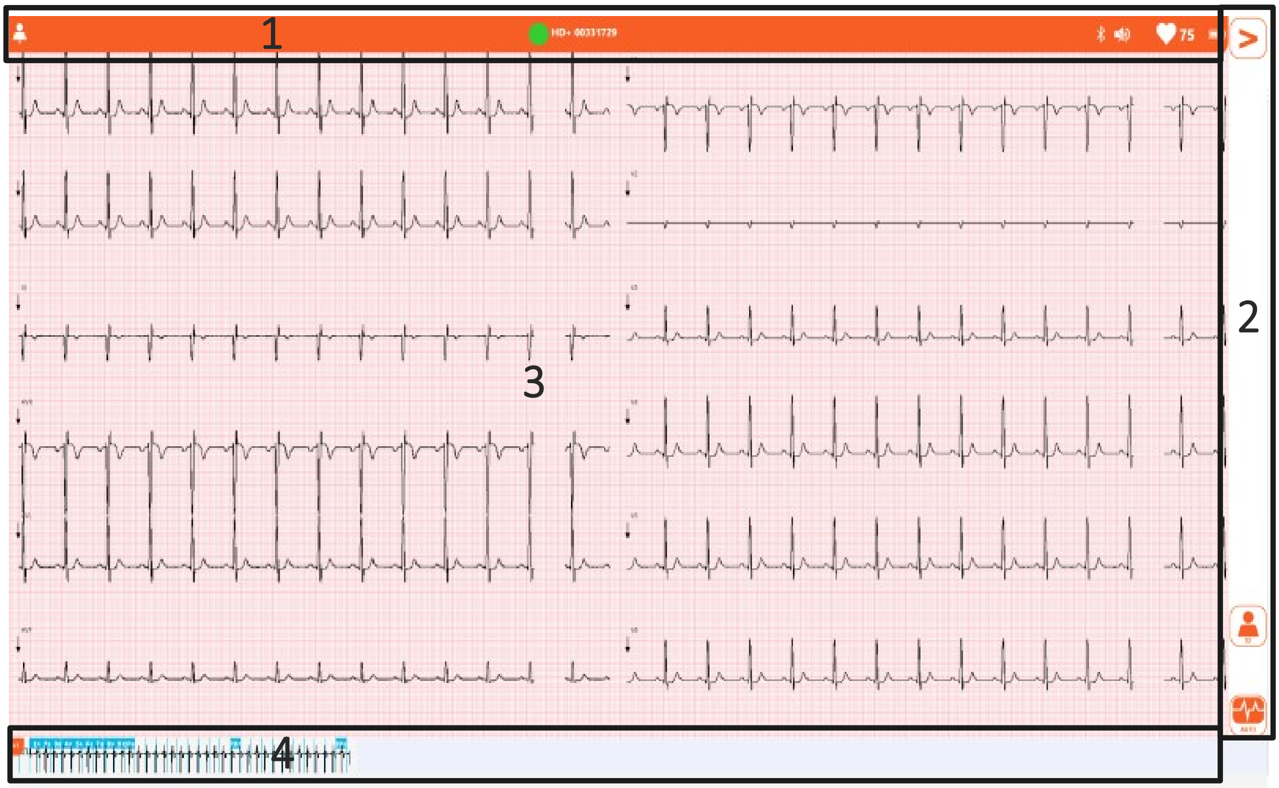
  the central area displays the contents of the window, for example the traces in the main window.

- Lower bar (4):

  at the bottom, depending on the window, it is used to show the continuous rhythm track or to show informational messages.



Main window with full User Interface



Main window with quick User Interface

4.5.1. Main buttons and icons

We list all commands available in touchECG.
For a detailed description of the commands and their functions, refer to the chapters describing the respective application windows.

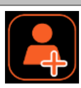
Top bar

Icon	Description
	Patient first and last name (if entered)
	HD+ connected / not connected
	QRS sound on / off
	Attention!
	Cardiac frequency (with value in bpm)
	HD+ battery level
	Number of tests to be sent
	Number of pages to print (only for *Windows* version)

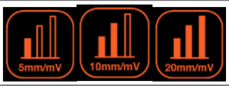
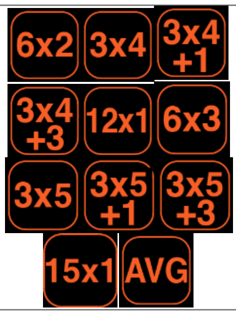
Side bar

C indicates the controls only featured in the full user interface
W indicates the controls only featured in the *Windows* version

Button	Version	Name	Description
Main window			
Menu 1			
		Open menu	Opens the secondary menu.
	C	Speed	Selects the trace speed 5, 10, 25, 50 mm/s
	C	Amplitude	Selects the trace amplitude 5, 10, 20 mm/mV

	C	Muscle filter	Selects the muscle filter 25 Hz, 40 Hz, 150 Hz, off ***Note****: The 150 Hz filter only works on 1000 Hz acquisitions and is only available in the* ***Windows*** *version.* ***Note****: The 25 Hz filter is stronger than the 40 Hz filter.* *Setting the filter to “off” disables muscle filtering of the traces.*
	C	Format	Selects the trace format 12 leads: 6+6, 3x1 (I-II-II), 3x1 (aVL, aVR, aVF), 3x1 (v1, v2, v3), 3x1 (v4, v5, v6), 12x1 15 leads: 6x3, 3x1 (I-II-II), 3x1 (aVL, aVR, aVF), 3x1 (v1, v2, v3), 3x1 (v4, v5, v6), 3x1 (e1, e2, e3), 12x1, 15x1
		Id	Opens the patient database for entering the patient details.
		Auto	Starts automatic 10s ECG acquisition
	C W	Review	Starts acquisition in review mode *Available in the* ***Windows*** *version only.*
	C W	Manual	Starts/stops manual mode acquisition *Available in the* ***Windows*** *version only.*
Menu 2			
		Close menu	Closes the secondary menu.
		Examination archive	Opens the examination archive
		Settings	Opens the settings window
	W	Launch App	Launches an external application (if so configured in the settings)
		Close	Closes touchECG
Patient window			
		New patient	Opens a new patient
		Search	Searches for a patient in the examinations archive

		Work list	Opens the work list window
		OK	Closes the patient window and saves the entered data.
		Back	Closes the patient window without saving the entered data.
Examinations archive window			
		Display	Displays the selected examination
		Transmit	Transmits the selected examination
		Transmit all	Transmits all untransmitted examinations
		Printing	Prints out the selected examination
		Send email	Sends the selected examination by email (as a PDF report)
		Remove	Deletes the selected examination
		Ok	Closes the examinations archive and saves the database data for the selected examination.
		Back	Closes the examinations archive without saving the database data for the selected examination.
Work list window			
		Update	Updates the work list by downloading it again.
		Delete	Deletes the order selected from the list (locally, not on the server from which the list has been downloaded).
		OK	Closes the window, saves the selection and loads the selected patient data.
		Back	Closes the window without saving the selection or loading the selected patient data.
Settings			
		Update license	Updates the program's active options.

		Save	Closes and saves the settings.
		Back	Closes without saving the settings
Examination preview window			
Menu 1			
		Open menu	Opens the secondary menu.
		Speed	Selects the trace speed 25, 50 mm/s
		Amplitude	Selects the trace amplitude 5, 10, 20 mm/mV
		Muscle filter	Selects the muscle filter 25 Hz, 40 Hz, 150 Hz, off ***Note****: The 150 Hz filter only works on 1000 Hz acquisitions and is only available in the* ***Windows*** *version.* ***Note****: The 25 Hz filter is stronger than the 40 Hz filter. Setting the filter to "off" disables muscle filtering of the traces.*
		Format	Selects the trace format 12 leads: 6x2, 3x4, 3x4+1, 3x4+3, 12x1 15 leads: 6x3, 3x5, 3x5+1, 3x5+3, 15x1
		Urgent	Turns the "urgent" status of the examination on/off.
		Transmit	Transmits the acquired examination
		Printing	Prints out the acquired examination
		Save / Update	Saves / updates the acquired examination
		Back	Closes the window and returns the main window in real time
Menu 2			

		Close menu	Closes the secondary menu.
	W	Calipers	Enables / disables the calipers tool. *Available in the **Windows** version only.*
		Send email	Sends the acquired examination by email (as a PDF report)
	W	Launch App	Launches an external application (configured in the settings) *Available in the **Windows** version only.*
Review window			
		Speed	Selects the trace speed 25, 50 mm/s
		Amplitude	Selects the trace amplitude 5, 10, 20 mm/mV
		Muscle filter	Selects the muscle filter 25 Hz, 40 Hz, 150Hz, off ***Note**: The 150 Hz filter works only on acquisitions at 1000 Hz. The 25 Hz filter is stronger than the 40 Hz filter.* *Setting the filter to "off" disables muscle filtering of the traces.*
		Format	Selects the trace format 12 leads: 6+6, 12x1, 1x1, 3x1 15 leads: 6+6, , 15x1, 1x1, 3x1
		Id	Opens the Patient database for entering the patient details.
		Print PDF	To save an exam in PDF format corresponding to the selected path or to the whole exam (if no area has been selected).
		Auto	Acquires a 10s ECG corresponding to the selected portion of the trace.
		Back	Closes the window and returns the main window in real time

5. PREPARATION FOR USE

5.1. Installing the software

5.1.1. Installation on *Windows*

Installation on Windows operating system may take place in the following ways:

- Insert the supplied USB stick into your computer (possibly provided as part of the system) and:
    - Double click on the ***setup.exe*** file, if an internet connection is available, or
    - Click on the ***touchECG.application*** file, if an internet connection is not available, or
    - Double-click on the ***touchECG_Setup.msi*** file.

5.1.2. Installation on *Android*

touchECG can be installed on the Android operating system simply by installing the APK (Android Application Kit) by running the touchECG.apk file.

For information on installing APK files, see the Google Android guide available in your country of origin.

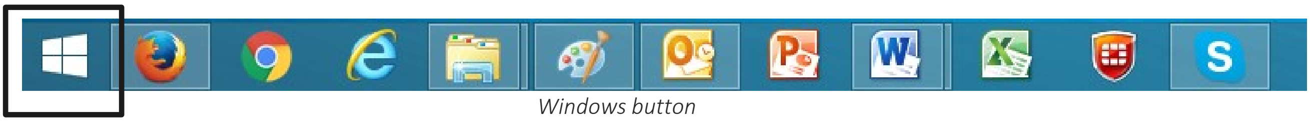
5.2. Launching touchECG

To launch the touchECG program, simply click on its desktop icon or in the applications list.

touchECG icon

Alternatively, you can select the program from the list of installed programs:

1. On ***Windows*** operating system, click on the **Windows** button in the application bar (see picture below) to open the main Windows menu.



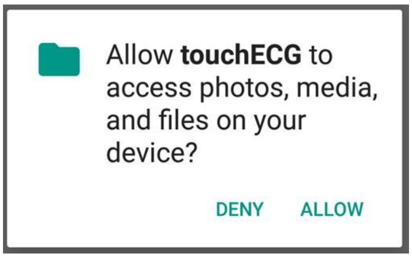
Windows button

On ***Android*** operating system, go the applications list.

2. In the list of applications (App) select touchECG.

Select touchECG

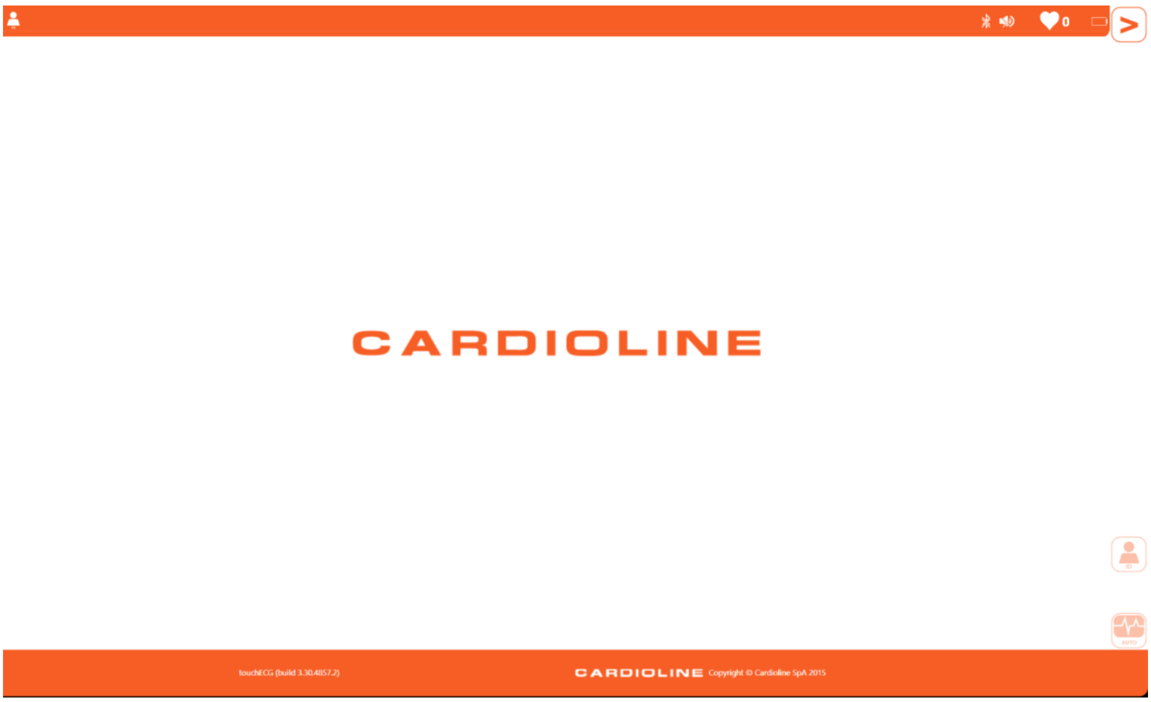
NOTE: *from TouchECG version 3.42 onwards, with Android operating system, the first time the program is launched a message is displayed, asking permission to access to the device's memory.*



Permission for access must be granted, otherwise the program ends.
Permissions may be entered manually from the menu Settings > Permissions > Memory permissions.

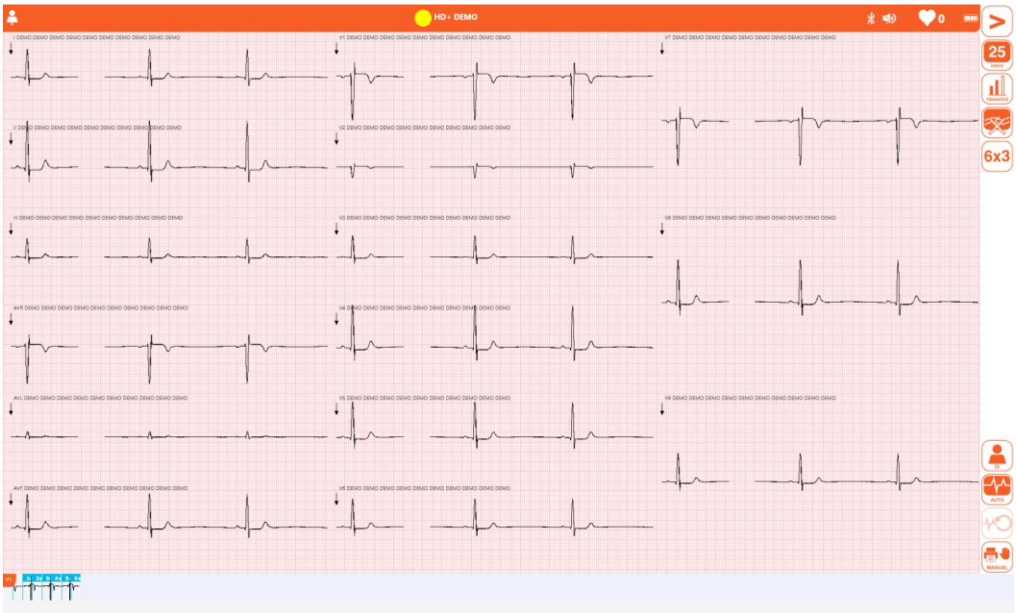
Note: *The program starts up with "Quick user interface" default*
Note: *if the "Boot Screen" mode is selected (see par. 5.2.1) TouchECG is started with a home screen that lets you enter the Ward ID.*

Once the program is launched the following window is displayed:



Start page

If no HD+ has been configured yet (see para. **Errore. L'origine riferimento non è stata trovata.**), TouchECG will start in DEMO mode showing a screen with sample traces. Video traces are marked with the word "DEMO," just as saved and printed tests.
Demo mode can be also enabled by selecting HD+ "Demo" in settings (see para. 8.2).

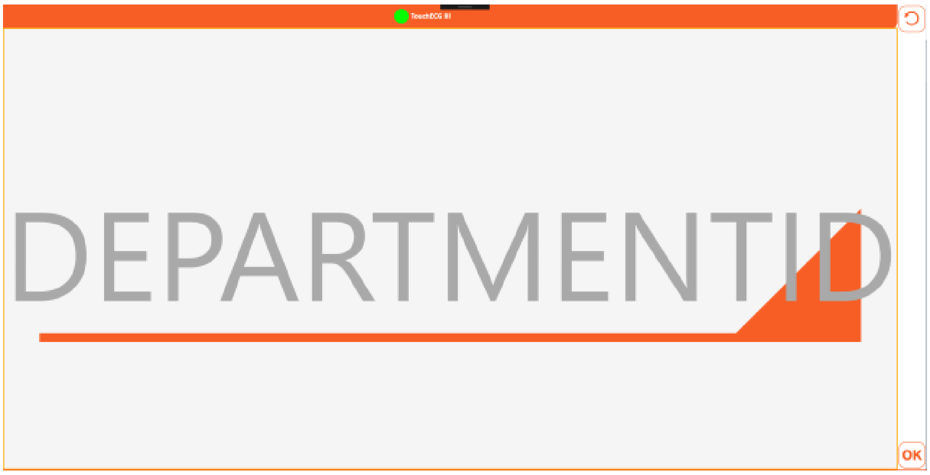


DEMO Mode

5.2.1. Start-up with manual entry of Ward Id

It is possible to set up TouchECG for it to start with a window to manually enter the Ward ID. This feature is useful if the device is often moved between wards with different Ward ID or in case of replacement equipment.

If the Boot Screen setting is enabled (see par. 8.2), upon start-up the window pictured below is displayed, in which the Ward ID may be entered.
The entered Ward ID is also saved in the program settings and becomes the default one.



Boot Screen for entering the Ward ID.

5.3. Connecting and configuring the HD+ or CLICKECG-HD acquisition unit

To acquire the ECH signal, the HD+ unit must be connected to the computer running touchECG (configured appropriately).

HD+ transmits via Bluetooth. Before making the connection, make sure that Bluetooth is available and enabled on the computer. If the computer does not have an integrated Bluetooth interface, use a USB Bluetooth adapter.

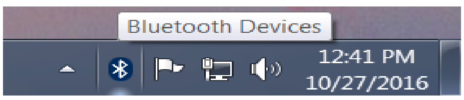
Note*: the HD+ unit must be enabled to connect to touchECG. If the HD+ unit was purchased previously to or separately from touchECG, check with Cardioline SpA that it is properly enabled.*

5.3.1. Connecting the HD+ acquisition unit to the Windows tablet/computer

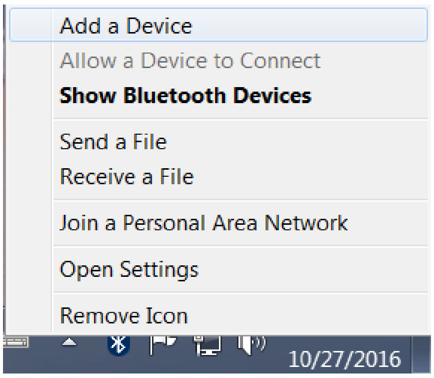
Power up the computer running touchECG and check that Bluetooth is available and enabled, then proceed as follows:

1. Insert the HD+ unit's batteries and turn it on.

2. On the computer/tablet device, click on the **Bluetooth devices** icon in the Windows application bar and select **Add a Bluetooth device**.



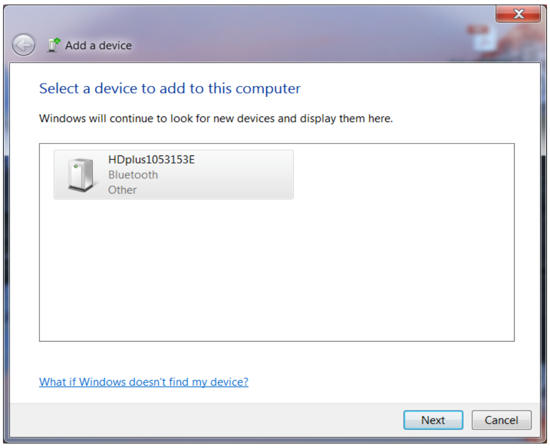
Bluetooth Devices Icon



Add a Bluetooth device window

3. In the **Manage Bluetooth devices** window, wait for the HD+ to be detected and added to the list of devices, with a name like **HDPlusxxxxxxxx** (where xxxxxxxx is the device's serial number) and the wording "Ready for pairing".

4. Click on the device and select **Pair**. Wait for the pairing procedure to complete, after which the HD+ will be connected to the system (the word "Connected" will display beneath the device).



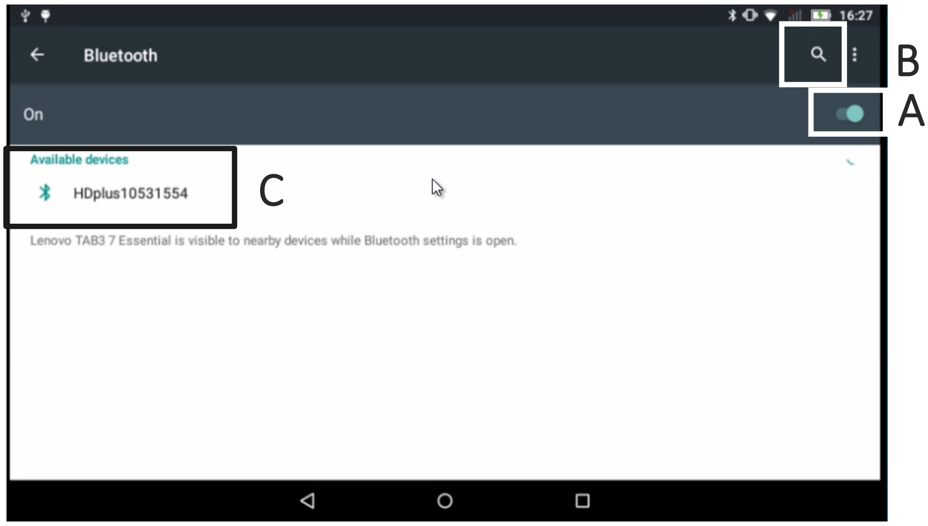
Manage Bluetooth devices window

5.3.2. Connecting the HD+ acquisition unit to the Android tablet/computer

Power up the device running touchECG and check that Bluetooth is available and enabled, then proceed as follows:

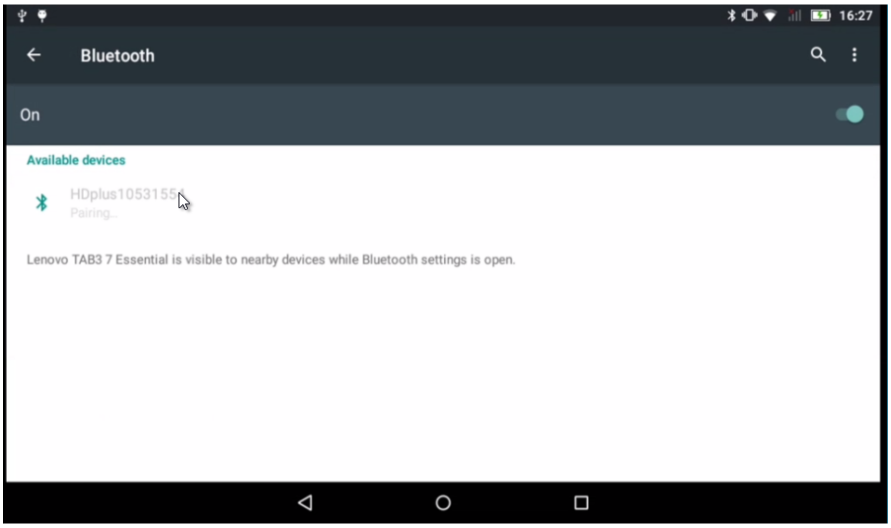
1. From the computer/tablet device, open **Settings** and go to the **Bluetooth** section.
2. Ensure Bluetooth connectivity is enabled (picture below **A**).
3. Insert the HD+ unit's batteries and turn it on.

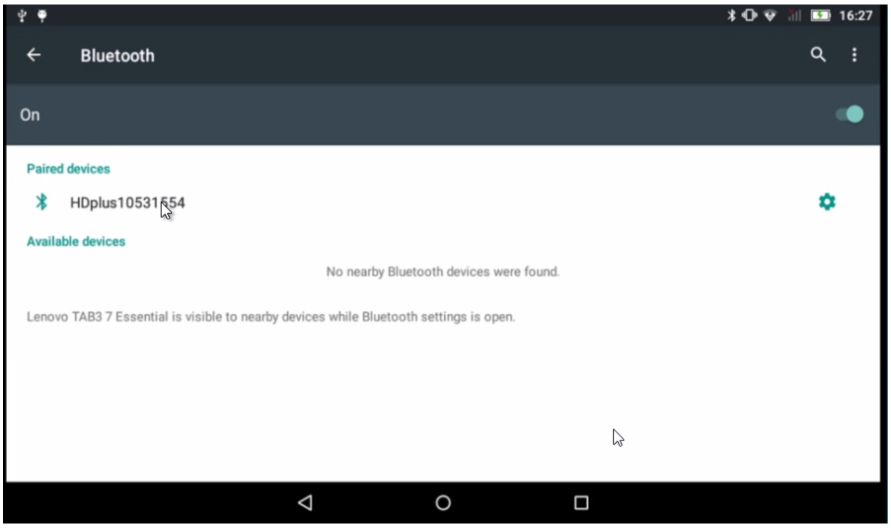
4. Click on the **Search** button (picture below **B**) to start HD+ search and wait for the HD+ device to be detected and added to the list of devices (picture below **C**), with a name like **HDPlusxxxxxxxx** (where xxxxxxxx is the device's serial number).



Bluetooth activation and HD+ search

5. Click on the device to perform the pairing. Wait for the pairing procedure to complete, after which the HD+ will be connected to the system.





HD+ Pairing

5.3.3. Configuring the HD+ acquisition device

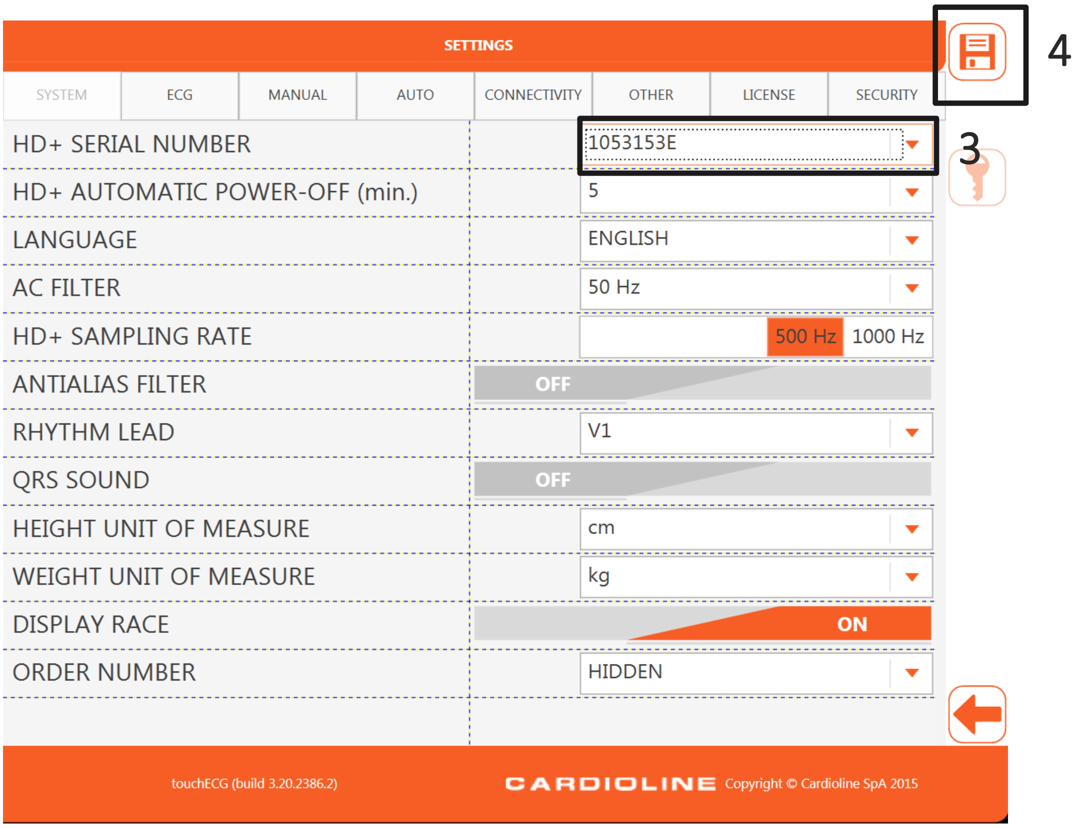
Once the HD+ unit is connected to the computer, you must configure touchECG to communicate with it.

Proceed as follows:

1. Launch touchECG (see para. 5.2).

2. Click on Open menu and then Settings (see para. 8.2)

3. Open the **System** tab and open the **HD+ serial number** dropdown menu which contains a list of all devices paired with the computer. Now select the serial number of the HD+ to want to connect to.
4. Click on **Save** to save this setting.



Settings window

Now touchECG will connect with the HD+ and the message “Searching for HD+” will display in the message area.
If the selected HD+ is not enabled to work with touchECG, the message “HD+ NON E' LICENZIATO PER IL TOUCHECG 3” (HD+ NOT LICENSED FOR TOUCHECG 3) will display. Contact Cardioline SpA to enable the acquisition unit.
If touchECG is unable to connect to the acquisition unit, because it is out of range or switched off, the message “HD+ not found" will display. Check that the acquisition unit is switched on and within range.

When the connection to the HD+ is successful, the message “HD+ found" displays in the message area.
Note*: the list of acquisition units paired with the computer always includes the "DEMO” device. Selecting this unit launches touchECG demo mode, which displays traces and runs the application even though no actual acquisition unit is connected. The demo mode trace was acquired with an ECG simulator and includes special signal conditions including disconnected leads and abnormal rhythms.*

5.4. Connecting and configuring the HD+ 12, HD+ 15, CLICKECG-HD 12 or CLICKECG-HD 15 acquisition unit

The new generation HD+ 12, HD+ 15, CLICKECG-HD 12 and CLICKECG-HD 15 acquisition units make for a faster and immediate connection to touchECG.

Connection via Bluetooth

If the HD+ device transmits via Bluetooth, before making the connection it is necessary:

- to make sure that the Bluetooth mode is active and available on the computer or that the HD+ DONGLE is installed;
- switch on the HD+ device.

Once the HD+ device is on, it suffices to launch touchECG and insert the HD+ serial number in the "System" section of the Settings.

Connection via USB

If the device transfers the data via USB, before making the connection it is necessary to connect the USB cable to the computer.
Once the device is connected via USB, it suffices to launch touchECG, which automatically detects the device connected and executes the connection.

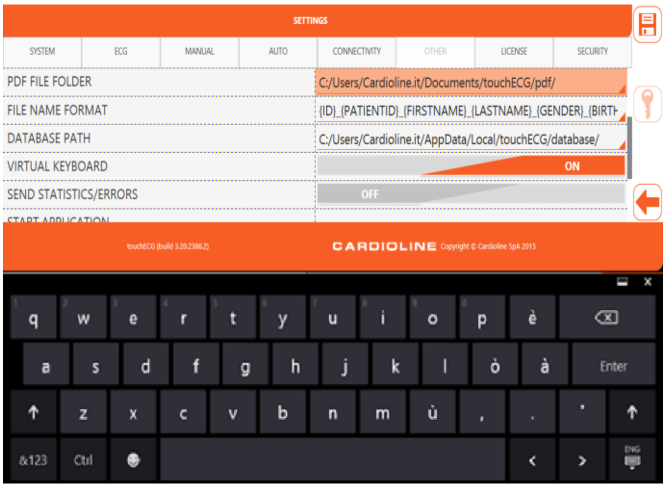
Note: the HD+ unit must be enabled to connect to touchECG. If the HD+ unit was purchased previously to or separately from touchECG, check with Cardioline SpA that it is properly enabled with the appropriate software license.

5.5. Virtual keyboard configuration (*Windows* version only)

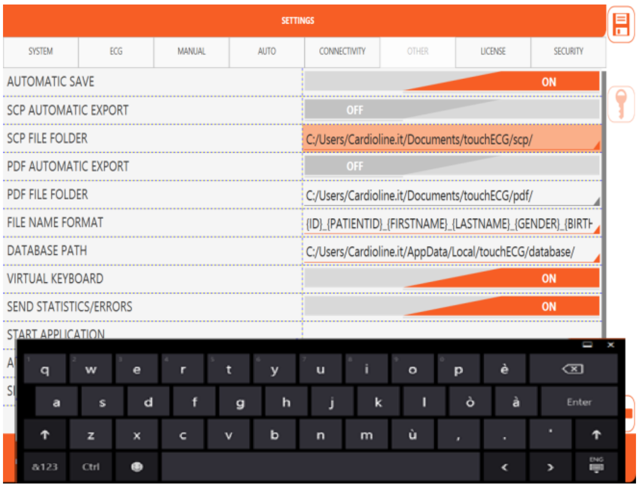
If not using a physical external keyboard but the virtual keyboard installed on the computer/tablet, it must be configured appropriately.
In particular, after making sure that the "Virtual keyboard" is set to On (see par. 8.2.6) it must be set to the "anchored" mode by clicking on symbol A in the picture and ensuring the keyboard does not overlap the application.

Virtual keyboard

CARDIOLINE



Virtual keyboard configured correctly



Virtual keyboard configured incorrectly

***Note**: if Windows is in tablet mode, it enables the virtual keyboard by default directly from the PC settings and, in this case, the "Virtual Keyboard" option in the touchECG must be **disabled**.*

5.6. Start from command line (*Windows* version only)

In some particular cases (for example in the event of touchECG software integrations with external systems) the program may need to be launched from a command line.
To do this, create a text file with ".cmd" or ".bat" extension containing the command:

For ClickOnce version:
`%AppData%"\Microsoft\Windows\Start Menu\Programs\Cardioline\touchECG\touchECG.appref-ms"`

Other commands can be added, each separated by a comma',' and enclosed in inverted commas "".
For Standard setup version:
`%AppData%"\Microsoft\Windows\Start Menu\Programs\Cardioline\touchECG\touchECG.lnk"`
Other commands can be added, each separated by a space ' '.

Command	Description	Syntax
-n	The application is launched in standard mode.	-n
-a	The application is launched by retrieving the plugin configured to receive the work list, loading the first patient and starting the acquisition in AUTO mode immediately.	-a
-g	The application is launched by retrieving the plug-in configured for the reception of the work list, loading the first patient and opening the display window in real time.	-g

Command	Description	Syntax
-w	The application starts by recalling the plugin configured to receive the work list, loading the first patient and opening the Patient window.	-w
-f	Path of the plug-in file for the reception of work lists	-f <PATHFILE> *E.g. Standard setup:* `-d C:\temp\wl\`
-d	Path of the folder where the received work list is saved	-d <PATH> *E.g. Standard setup:* `-d C:\temp\gdt\input\`
-pg	Name of the dl plug-in file for the reception of work lists	-pg <NAME> *E.g. Standard setup:* `-pg Cardioline.TouchPlugWebApp`
-ps	Name of the dl plug-in file for the transmission of examinations	-ps <NAME> *E.g. Standard setup:* `-ps Cardioline.TouchPlugWebApp`
-u	Username for the plug-in	-u <USERNAME> *Es. Standard setup:* `-u uploader`
-p	Password for plug-in	-p <PASSWORD> *E.g. Standard setup:* `-p Cardioline`
-l	Plug-in url address or connection string	-l <URL> *Es. Standard setup:* `-l http://213.209.216.187:9091/ecgwebapp/rest`
-c	Allows you to use a custom user configuration	-c <PATHFILE> *E.g. Standard setup:* `-c C:\temp\user.config`
-w	It allows you to position the application in the management of the working list as soon as the connection to the device is made	*Es. Standard setup:* `-w`

Where:
<NAME> = name of the dl plug-in file written without the "dl" extension
<PATH> = path written with "\" at the end
<PATHFILE> = path and file name
<URL> = web address/url
Example of a complete string for the Standard setup version:

`%AppData%"\Microsoft\Windows\Start Menu\Programs\Cardioline\touchECG\touchECG.lnk"`
-pg -Cardioline.TouchPlugWebApp -a

***Note**: there must ALWAYS be a space between the topics.*

5.7.1. Windows version

The software can be updated by installing the new version of the touchECG setup.
The updated manuals are included in the installation media provided.

5.7.2. Android version

The software is updated by running the new touchECG APK file.
The updated manuals are included in the installation media provided.

5.8. System installation

As indicated in par. 4.4 the device can be supplied as a touchECG System, consisting of the touchECG software, the HD + acquisition unit, the Cardioline HD + Bluetooth Dongle, the computer, the trolley and the other listed devices.
Each device has its instructions for use. Refer to them for the installation of the individual devices (e.g. of the trolley) and to the touchECG System manual for indications relating to the system.

6. EXECUTION OF THE EXAMINATION

6.1. General procedure

To acquire an ECG proceed as follows:

1. Start touchECG (see para.5.2)
2. Prepare and connect the patient (para. **Errore. L'origine riferimento non è stata trovata.** and 6.3)
3. Turn on the HD+ acquisition device by pressing the central button (see the relevant manual)
4. Check the quality of the traces on the display and make sure there are no error messages (para. 6.4).
5. Complete the patient details, if necessary (para. 6.5)
6. Acquire the examination (see para. 6.6) by pressing button:
   - **Auto** or the central button on the HD+ acquisition device for an automatic ECG acquisition,
   - **Review** (only available in the ***Windows*** version) to review the whole acquired tracing
   - **Manual** (only available in the ***Windows*** version) for manual ECG acquisition.

NOTES: If the work flow allows it, it is good practice to connect the patient to the device and to enter his/her ID data before recording a tracing. This minimises artefacts on the tracings introduced during connection of the patient and positioning of electrodes.

6.2. Preparing the patient

Ensure that the patient fully understands the procedure and knows what to expect before connecting the electrodes.

- Privacy is very important to allow the patient to be relaxed.
- Reassure the patient that the procedure is painless, and that he/she will only feel the electrodes on the skin.
- Make sure that the patient is relaxed and in a comfortable position. If the table is narrow, insert the hands of the patient under the buttocks to ensure the muscles are relaxed.
- Once the electrodes are connected, ask the patient to remain still and not to talk. Explain that this is important to ensure a good ECG acquisition.

It is important that the patient's skin be perfectly clean. There is a natural electrical resistance on the surface the skin, generated by sources such as hair, sebum, and dry or dead skin. Adequately prepare the skin to minimise the aforementioned effects and optimise the quality of the ECG signal.
To prepare the skin:

- If necessary, shave the skin area where the electrode must be applied.

- Wash the area with hot soapy water.
- Dry the skin vigorously with an abrasive pad, such as a 2x2 or 4x4 gauze, to remove dead skin cells and fat.

***Note**: Pay attention not to cause abrasions, discomfort or bruises on the skin. Always observe the utmost clinical discretion when preparing the patient.*

6.3. Connecting the patient

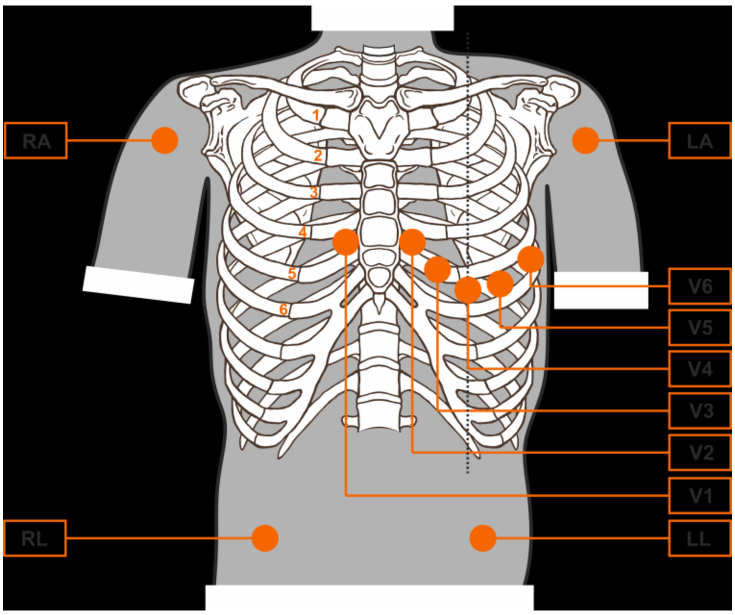
It is important to position the electrodes properly in order to acquire a good electrocardiographic signal. Indeed low impedance ensures a better waveform, reducing noise. Quality electrodes should be used.

Connect the electrodes as follows:

1. Expose the arms and legs of the patient to connect the relevant leads.
2. Position the electrodes on the flat and fleshy parts of the arms and legs.
3. If a limb is not available, position the electrodes on a blood-supplied stump.
4. Attach the electrodes to the skin as indicated in the following figure. Test the correct adherence, and therefore the good contact, by pulling the electrode. If the electrode moves freely, replace it. If the electrode does not move easily, a good electrical contact is ensured.

6.3.1. 10-wire cable connection (for acquisition of 12-lead ECG)

Reference figure for the connection to the patient



***Note**: It is important to locate the fourth intercostal space for an accurate positioning and monitoring of the precordial leads. It is possible to locate the fourth intercostal space starting from the first intercostal space. Given the variable conformation of the patient, palpating the first intercostal space accurately can be difficult. Therefore, it is advisable to locate the second intercostal space by first palpating the small bone protrusion known as the Angle of Louis, formed by the junction of the manubrium and the body of the sternum. This protrusion of the sternum identifies the junction point of the second rib, and the space immediately below it corresponds to the second intercostal space. Palpate and count following the trunk until the fourth intercostal space is located.*

Reference table for the connection to the patient

IEC electrode			AAMI electrode			Position
C1	C1	Red	V1	V1	Red	Fourth intercostal space to the right of the sternum.
C2	C2	Yellow	V2	V2	Yellow	Fourth intercostal space to the left of the sternum.
C3	C3	Green	V3	V3	Green	Midway position between electrodes V2/C2 and V4/C4.
C 4	C4	Brown	V4	V4	Blue	Fifth intercostal space on the left of the midclavicular line.
C5	C5	Black	V5	V5	Orange	Between electrodes V4 and V6.
C6	C6	Violet	V6	V6	Violet	Level with electrode V4 at left midaxillary line.
L	L	Yellow	AL	LA	Black	On the deltoid muscle, the forearm or left wrist.
R	R	Red	RA	RA	White	On the deltoid muscle, the forearm or right wrist.
F	F	Green	LL	LL	Red	On the thigh or the left ankle.
N	N	Black	RL	RL	Green	On the thigh or the right ankle.

6.3.2. 13-wire cable connection (for acquisition of 15-lead ECG)

The 13-wire cable includes the 10 standards leads labelled and coloured as mentioned in the table above and 3 "Extra" leads, identified with the labels E1, E2 and E3.

IEC lead			AAMI lead		
E1	E1	White	E1	E1	Brown
E2	E2	White	E2	E1	Brown
E3	E3	White	E3	E1	Brown

The extra leads are used as additional single-pole leads with regard to the standard precordial leads (V1-V6). The E1, E2 and E3 electrodes can be placed at the extended precordial leads described in the figure below.

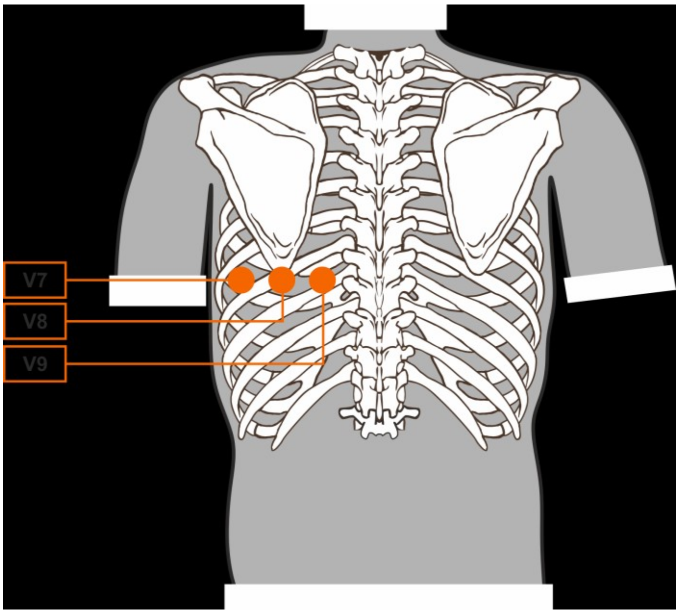
The most common placement combinations of extended leads are:

- **Posterior Placement**: E1 placed at position V7, E2 placed at position V8, E3 placed at position V9
- **Right Placement**: E1 placed at position V3R, E2 placed at position V4R and E3 placed at position V6R
- **Paediatric Placement**: E1 placed at position V3R, E2 placed at position V4R, E3 placed at position V7

NOTE: the choice of placement used depends on the clinical investigation performed.

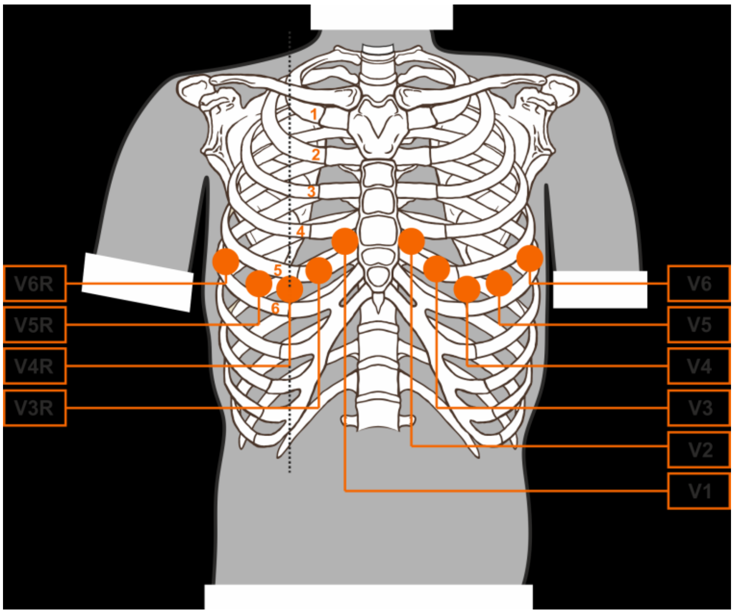
The figures below show the most common placement of extended leads.

Reference figure for the connection to the patient of the Posterior Leads



- **V7**: left posterior axillary line, level with lead V6
- **V8**: tip of the left scapula, level with V7
- **V9**: left paraspinal line, level with V8

Reference figure for the connection to the patient of the Right Leads



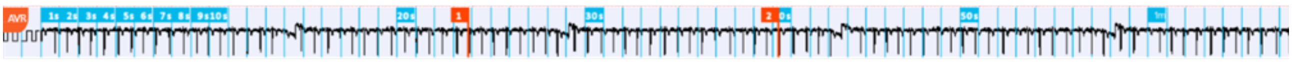
- **V3R**: Fifth rib between V1 and V4R (same level as electrode V3)

- **V4R**: Fifth intercostal space on the right midclavicular line (same level as V4)
- **V5R**: Fifth intercostal space, halfway between V6R and V4R (same level as V5)
- **V6R**: Fifth intercostal line on the right mid-axillary line (same level as V6)

6.4. Viewing the ECG

If touchECG is connected to an HD+ unit, turned on and connected to a patient, the touchECG start window is the real time signal display page. The message "Real-time start" displays in the message area.
The centre of the screen is dedicated to displaying the traces, and the display mode can be set with the buttons in the side bar, as described below (para. 6.4.1).

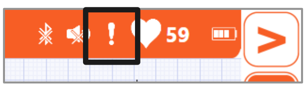
Since touchECG acquires and temporarily stores the entire trace acquired when it is launched and connected to an active HD+, the bottom of the screen (**A**) displays a continuous trace which corresponds to the entire trace acquired up to that time.
The lead displayed in the continuous trace can be modified from the Settings menu (see par. 8.2) or by clicking on the continuous trace itself.
On the blue continuous trace there are time markers identifying the duration of the acquired signal, identified at 1s intervals for the first 10s, and 10s intervals for the first minute, and at 1 min intervals thereafter.
Any bookmarks (see par. 6.4.3) are marked on the continuous trace as vertical orange lines identified by a tag indicating the progressive number.



Rhythm lead

The top bar displays a set of information icons whereas the side bar contains the buttons that correspond to the functions available which, depending on whether the "Quick user interface" setting is active or not, allow you to modify the track display mode, acquire a test, access the settings menu, and so on.

In case of signal loss caused by Bluetooth problems, the "Attention" icon appears in the top bar. If more than 4 ms of signal are lost, the message "Weak signal" also appears.
The signal can be acquired just the same, but if the acquired signal were to show consecutive signal losses of at least 100 ms or an overall loss of 1 s, the test would be classified as low quality (see par. 6.7).



Attention icon

***Note**: By default the touchECG is set to "Quick user interface". Should you wish the "Full user interface" it must be selected from the settings (see par. 8.2.6).*

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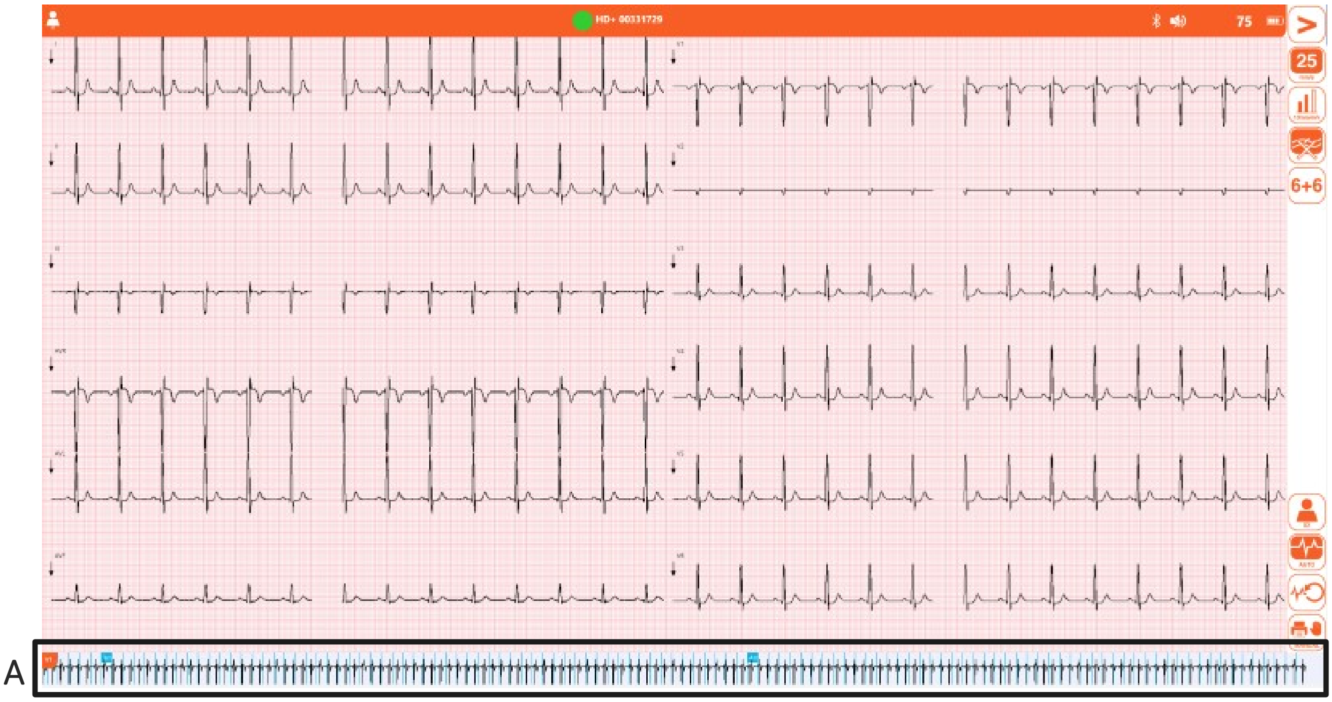
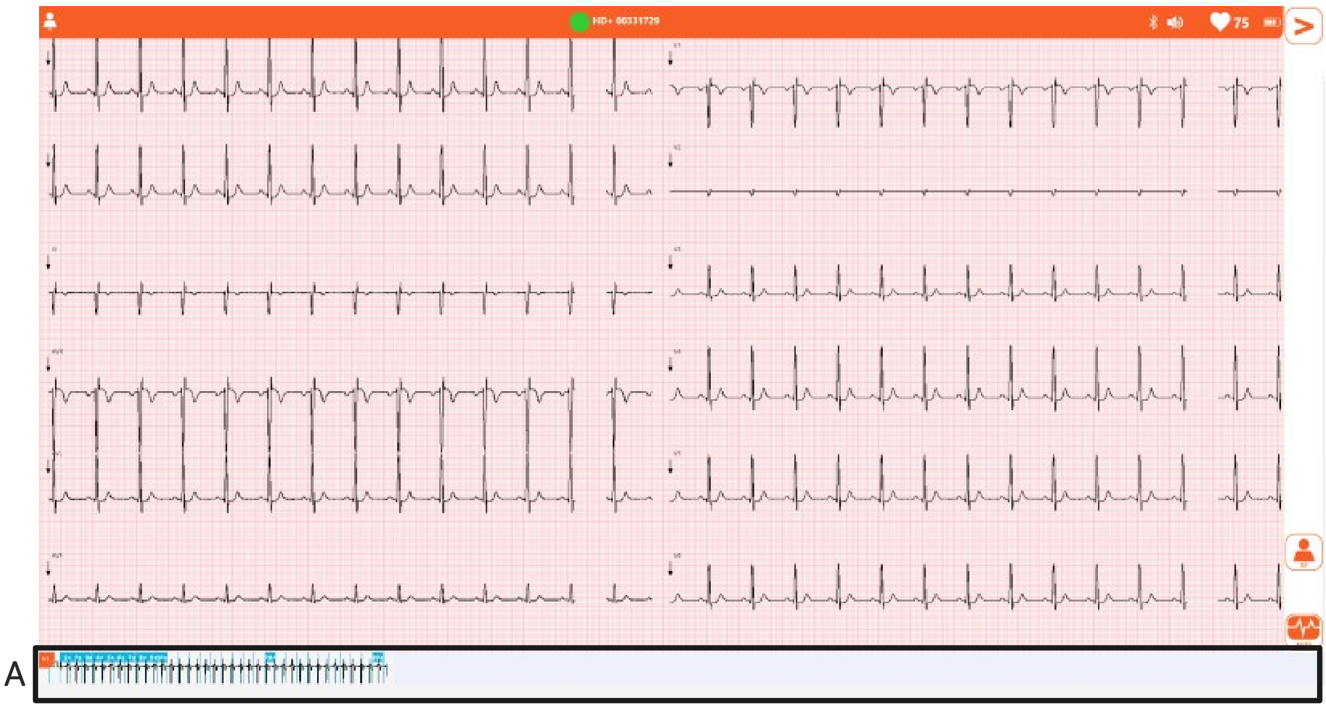


Real time display window with Full user interface



Real time display window with Quick user interface

Information icons – top bar

- **Patient ID** — Followed by the patient's first and last name, if entered, to identify the patient. It also enables to enter the patient's data (Par. 6.5).
- **HD+ connected / not connected** — Indicates whether the HD+ device is connected via Bluetooth.
- **Heartbeat sound on / off** — Indicates whether the heartbeat sound is on or off. If it is on, the unit beeps for each heartbeat.
- **Attention!** — Indicates that some leads are disconnected or that there is signal loss.
- **Cardiac frequency** — Followed by the number of beats (bpm).
- **HD+ battery level** — Indicates the HD+ unit's battery level.

Available buttons – side bar:

C indicates the controls only featured in the full user interface
W indicates the controls only featured in the *Windows* version

Menu 1

- **Open menu** — Opens secondary menu, Menu 2.
- C **Speed** — Selects the trace speed
5, 10, 25, 50 mm/s
- C **Amplitude** — Selects the trace amplitude
5, 10, 20 mm/mV
- C **Muscle filter** — Selects the muscle filter
25 Hz, 40 Hz, 150 Hz, off
*Note: The 150 Hz filter only works on 1000 Hz acquisitions and is only available in the **Windows** version.*
Note: The 25 Hz filter is stronger than the 40 Hz filter. Setting the filter to "off" disables muscle filtering of the traces.
- C **Format** — Selects the trace format
12 leads: 6+6, 3x1 (I-II-II), 3x1 (aVL, aVR, aVF), 3x1 (v1, v2, v3), 3x1 (v4, v5, v6), 12x1
15 leads: 6x3, 3x1 (I-II-II), 3x1 (aVL, aVR, aVF), 3x1 (v1, v2, v3), 3x1 (v4, v5, v6), 3x1 (E1, E2, E3), 15x1

	Id	Opens the patient window for entering the patient details (para. 6.5)
	Auto	Starts automatic 10s ECG acquisition (para. 6.6.1)
C W	Review	Starts acquisition in review mode (para. 6.6.2) *Available in the **Windows** version only.*
	Manual	Starts/stops acquisition in manual mode (para. 6.6.3)

Menu 2

	Close menu	Closes the secondary menu and returns to Menu 1.
	Examination archive	Opens the examination archive (para. 6.8)
	Settings	Opens the settings window (para. 7)
W	Launch App	Launches an external application (configured in the settings)
	Close	Closes touchECG.

6.4.1. Changing the trace display mode

If full user interface mode is enabled ("Quick user interface" setting = Off) it is possible to change the speed, amplitude, format and muscle filter applied to the tracings by clicking on the corresponding buttons.
Each button cycles through its available values, and applies them to the traces. For example, the speed button can take the values 5, 10, 25 and 50 mm/s, and each time you click the button it will move to the next value in the sequence.
The button itself displays the value.

We list the values of the parameters and the button labels below.

Tracings display format:

Speed

Button	Value
5 mm/s	5 mm/s
10 mm/s	10 mm/s
25 mm/s	25 mm/s

- 50 mm/s

Amplitude

- 5 mm/mV
- 10 mm/mV
- 20 mm/mV

Muscle filter

- 25 Hz
- 40 Hz
- 150 Hz

  Note: The 150 Hz filter only works on 1000 Hz acquisitions and is only available in the ***Windows*** *version.*
- off

Note: the 25 Hz filter is stronger than the 40 Hz filter. Setting the filter to "off" disables muscle filtering of the traces.

Format

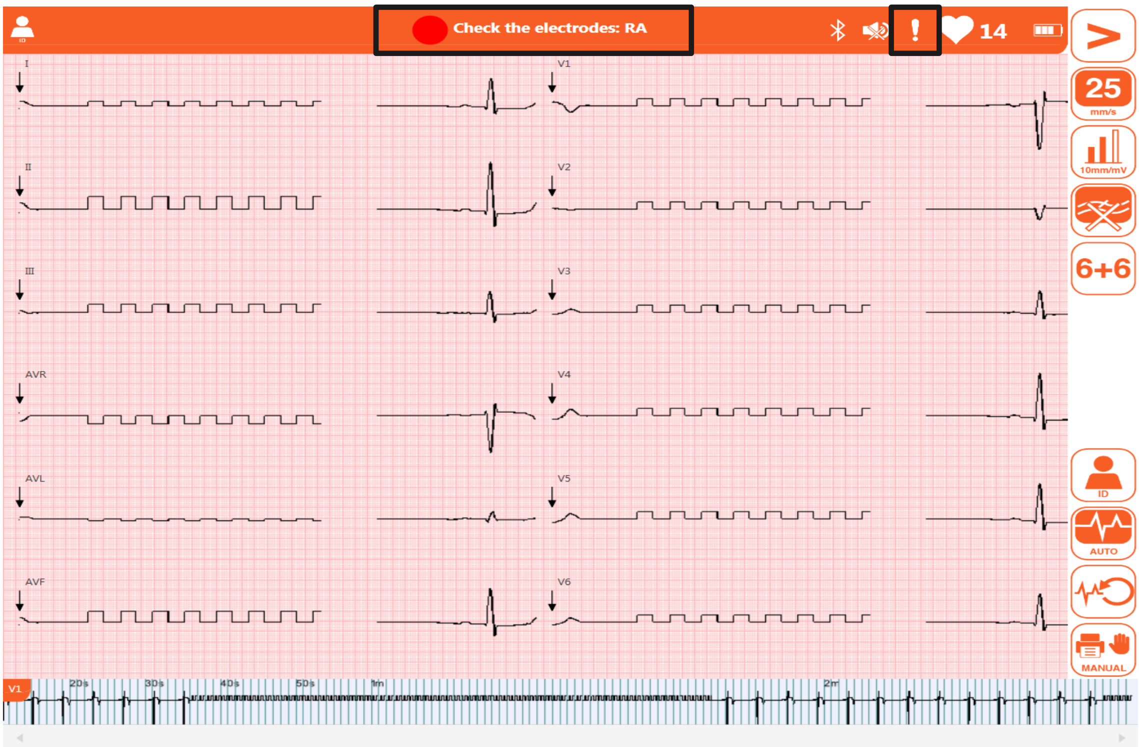
- 6+6
- 3x1 (I-II-II)
- 3x1 (aVL, aVR, aVF)
- 3x1 (v1, v2, v3)
- 3x1 (v4, v5, v6)
- 3x1 (E1, E2, E3) Only with HD+ 15 and CLICKECG-HD 15
- 12x1
- 15x1 Only with HD+ 15 and CLICKECG-HD 15

6.4.2. Leads disconnected

If any electrodes are disconnected, the message "Lead disconnected" displays in the message area, followed by the list of leads in question, and the icon **Attention** displays in the top bar.

The waveforms of the disconnected leads are displayed as square waves (see below).
If all the leads are disconnected, all ECG traces display as square waves and the message "Lead disconnected: all" displays. In the same way, if the disconnected lead is N/RL, the message "Lead disconnected: all" displays and all leads display as square waves.

If the patient cable is connected from the connector on the HD+ unit, the ECG traces display as flat lines.



Real time display window with disconnected lead.

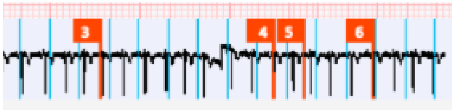
6.4.3. Bookmark

The real time display window allows you to assign bookmarks to events and interesting points of the signal by clicking with the mouse (or with your finger if using a touchscreen device) in the trace area at the time of interest.

The selected event is marked with vertical orange line identified by a progressive number visible in the continuous trace at the bottom of the screen.
If a test is acquired, the bookmark is also shown in the ECG test.

In the **Windows** version the bookmark is included in the ECG test acquired in Review mode as well as in the test acquired in automatic acquisition mode (see par. 6.6.1 and 6.6.2). In the Android version, which does not include the Review feature, the bookmark is shown in the test acquired in automatic mode (see para. 6.6.1 and 6.6.2).

You can set touchECG to enter bookmarks automatically as set intervals. The "Automatic bookmark" function in Settings (para. 8.2) enables you to select the interval in minutes between bookmarks.
The "Save test on Bookmark" function, on the other hand, automatically creates a 10 s test on every bookmark. If the automatic print function is active, the created test is also printed out.



Bookmarks highlighted on the continuous trace

6.5. Entering patient data

You can enter patient details before acquiring an ECG examination by pressing the **Id** button in the real time window or the **Id** field in the top bar, which opens the Patient window.

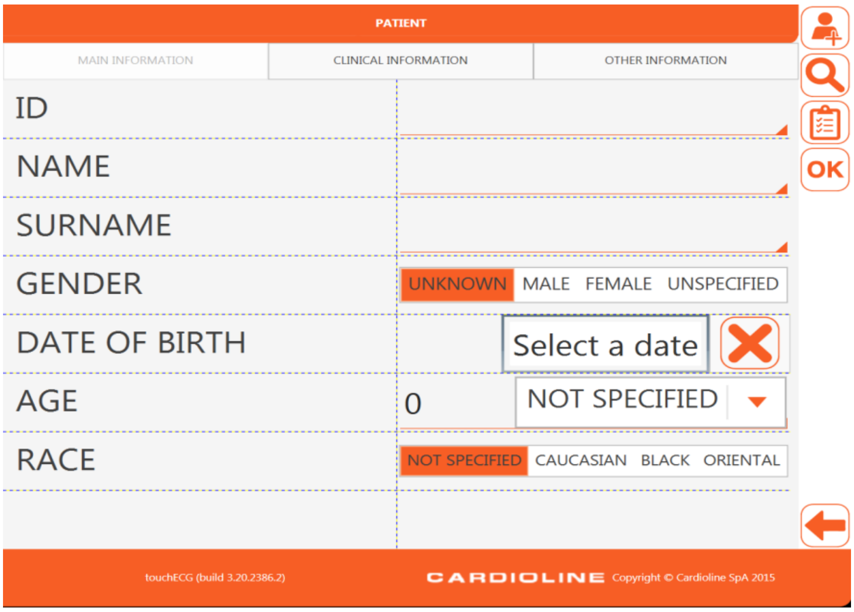
Id *button*

The data can be entered manually or automatically by selecting the patient from a work list or the examinations archive.

6.5.1. Patient window

The window is organised into 3 sections, each of which has a number of entry fields.
The sections are:

- **Main information**: the basic patient details
- **Clinical information**: weight, height, therapy, etc.
- **Other information**: contact details



Patient window

We describe the fields and buttons in the side bar below.

Fields:

Main information

▪ ID	the patient's identification code	alphanumeric text field
▪ Name	the patient's name	alphanumeric text field
▪ Surname	the patient's surname	alphanumeric text field
▪ Gender	the patient's gender	multiple-choice button: unknown / male / female / not specified
▪ Date of birth	date of birth of the patient	Numerical field (use '/', '-' or 'space' to enter the date). The **X** button allows the date entered to be deleted.
▪ Age	the patient's age	Free numerical field or calculated automatically if a valid date of birth is entered. The age can be expressed in years, months, weeks or days.

Note: If the age is not entered before requiring an ECG, it will be interpreted by default relative to a 50-year-old male. The interpretation text will include the wording "Interpretation done without knowledge of the patient's age".

Note: If the age zero (0) is used, the interpretation will use the default

	settings for a 50 year old male. The interpretation text will include the wording "Interpretation done without knowledge of the patient's age".	
▪ Race	the patient's race	Multiple-choice button: not specified / caucasian / black / oriental. The field will display or not according to the chosen setting. If it is not visible, the default value is "not specified".

Clinical information

▪ Weight and unit of measurement	the patient's weight	Numerical field. The units can be chosen from kg/g/lb/oz
▪ Height and unit of measurement	the patient's height	Numerical field. The units can be chosen from cm/in/mm
▪ Systolic	the patient's systolic pressure, in mmHg	numerical field
▪ Diastolic	the patient's diastolic pressure, in mmHg	numerical field
▪ SpO2	the patient's oxygen saturation	numerical field
▪ Pharmacological therapy	the patient's pharmacological therapy	alphanumeric text field
▪ Technician	the technician administering the examination	alphanumeric text field
▪ Reasons for study	diagnosis already confirmed or suspected, or main symptom	alphanumeric text field

Other information

▪ E-mail address	the patient's email address	alphanumeric text field
▪ Address	the patient's physical address	alphanumeric text field
▪ Order number	the reservation number for the examination (visible or editable depending on the selected configuration)	alphanumeric text field

Buttons – side bar:

- New patient — Deletes any existing patient data and creates a new patient
- Search — Finds a patient in the examinations archive (para. 6.5.3)
- Work list — Finds a patient in a work list (par. 6.5.4)
- OK — Closes the patient window and saves the entered data.
- Back — Closes the patient window without saving the entered data.

Note*: Make sure that the data have been deleted and re-entered before running an ECG for a new patient; the patient details are not reset automatically after the examination. The patient details are only deleted automatically when the device is switched off or you press* ***New patient****.*

6.5.2. Manually entering the patient's data

The data can be entered manually into the fields in the Patient window.
Once entered, you can save the details by clicking **OK** or abort the operation with **Back**.

6.5.3. Entering patient details from the Examinations Archive

Click on **Search** in the patient window to search for a patient in the local archive; select a patient and load his details.

Search *button*

It is possible to restrict the search by filling in all or part of the Id, Name, Surname fields of the Patient window before clicking the **Search** button.
touchECG will only search in the local archive for tests in which the patient records match the entered data, starting from the ID and then moving on to the surname and lastly, the name.

Alternatively, you can launch a search in the Examinations Archive by entering the word you are looking for in the **Search** field. touchECG will search the local archive only for examinations in which the word in question is found in the Id or Name/Surname fields.

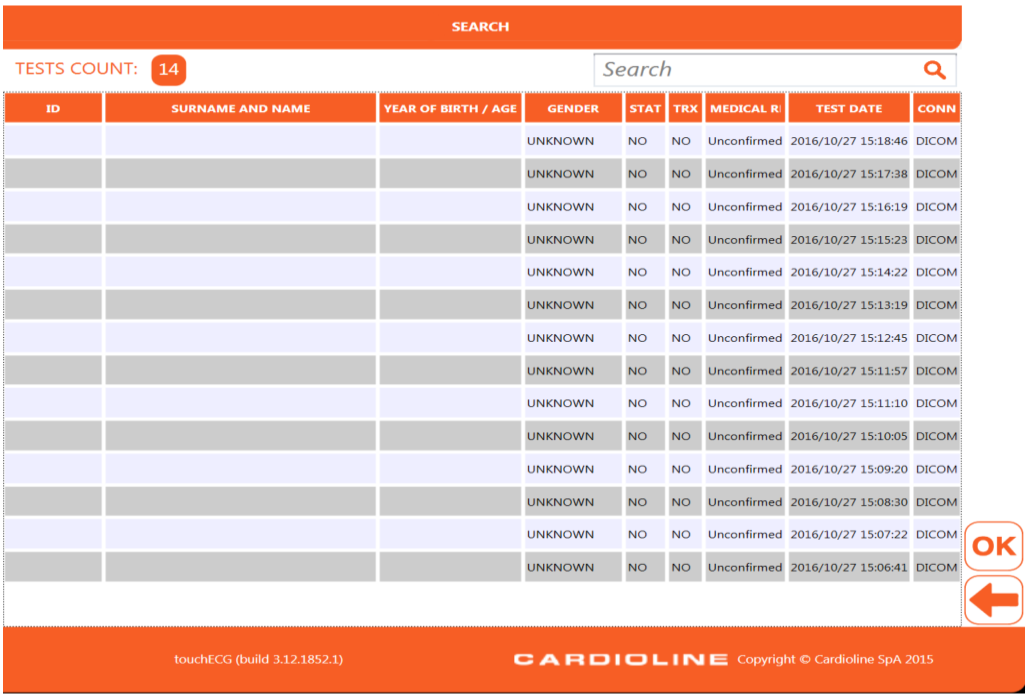
The following fields are displayed in order for each examination:

- Id: the patient's identification code
- Name and surname: the patient's name and surname

- Year of birth / age: the patient's year of birth and age
- Gender: the patient's gender
- Stat: the examination's urgent flag (Yes = urgent examination)
- Trx: the examination's transmitted flag (Yes = examination transmitted)
- Report: indicates whether the interpretation of the examination has been confirmed by a doctor or not
- Date of examination: date and time of the examination
- Connectivity: connectivity option associated with the examination (linked to the HD+ unit used to acquire the ECG, as described in par. 10).

Click on the column labels (e.g. "ID" or "Name and Surname) to order the list in increasing/decreasing order by the field in question.

You can select an examination by clicking on its line in the list and use **OK** and **Back** to save the data or abort the operation.



ID	SURNAME AND NAME	YEAR OF BIRTH / AGE	GENDER	STAT	TRX	MEDICAL R	TEST DATE	CONN
			UNKNOWN	NO	NO	Unconfirmed	2016/10/27 15:18:46	DICOM
			UNKNOWN	NO	NO	Unconfirmed	2016/10/27 15:17:38	DICOM
			UNKNOWN	NO	NO	Unconfirmed	2016/10/27 15:16:19	DICOM
			UNKNOWN	NO	NO	Unconfirmed	2016/10/27 15:15:23	DICOM
			UNKNOWN	NO	NO	Unconfirmed	2016/10/27 15:14:22	DICOM
			UNKNOWN	NO	NO	Unconfirmed	2016/10/27 15:13:19	DICOM
			UNKNOWN	NO	NO	Unconfirmed	2016/10/27 15:12:45	DICOM
			UNKNOWN	NO	NO	Unconfirmed	2016/10/27 15:11:57	DICOM
			UNKNOWN	NO	NO	Unconfirmed	2016/10/27 15:11:10	DICOM
			UNKNOWN	NO	NO	Unconfirmed	2016/10/27 15:10:05	DICOM
			UNKNOWN	NO	NO	Unconfirmed	2016/10/27 15:09:20	DICOM
			UNKNOWN	NO	NO	Unconfirmed	2016/10/27 15:08:30	DICOM
			UNKNOWN	NO	NO	Unconfirmed	2016/10/27 15:07:22	DICOM
			UNKNOWN	NO	NO	Unconfirmed	2016/10/27 15:06:41	DICOM

Local archive search window

Buttons – side bar:

OK		Closes the examinations archive and saves the database data for the selected examination.
Back		Closes the examinations archive without saving the database data for the selected examination.

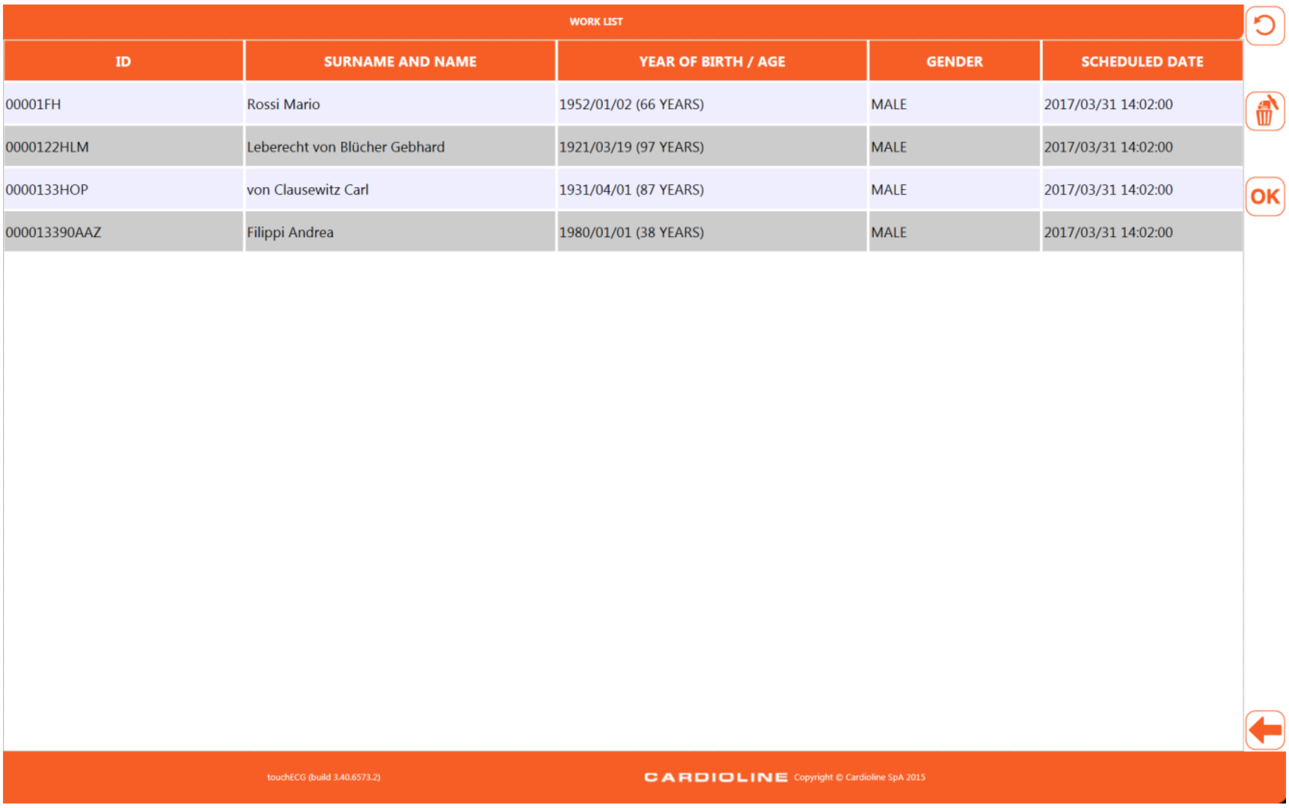
6.5.4. Entering patient details from a work list

The connectivity function (if available on the support device) enables you to load the patient details automatically by selecting them from a work list. For details on configuring the ECG unit, see par. 7.

Click on **Work List** in the patient window to download and display a work list.

Work list button

You can select an order from the displayed work list by clicking on the line in question, and then select it (**OK**), delete (**Delete**) or abort (**Back**).
You can also update the work list by pressing **Update** to download it again.



Work list

Buttons – side bar:

-  Update — Update the work list, deleting the current list and downloading it again.
-  Delete — Deletes the order selected from the list (locally, not on the server from which the list has been downloaded).

- 

OK — Closes the window, saves the selection and loads the selected patient data.
-  Back — Closes the window without saving the selection or loading the selected patient data.

6.6. Acquisition of an ECG examination

The ECG can be acquired in three ways:

- Automatic mode (**Auto** button or middle button on the HD+): acquires a 10s ECG which may be saved and printed out as an examination.
- In review mode (**Review** button): displays the trace acquired so far; you can select 10s portions of this and save and print them out as examinations. This function is not available if the “Quick user interface” setting is enabled. This button is only enabled after at least 10 seconds of acquisition.
- Manually (**Manual** button): prints a continuous ECG of variable duration (saving is not available). This function is not available if the “Quick user interface” setting is enabled.

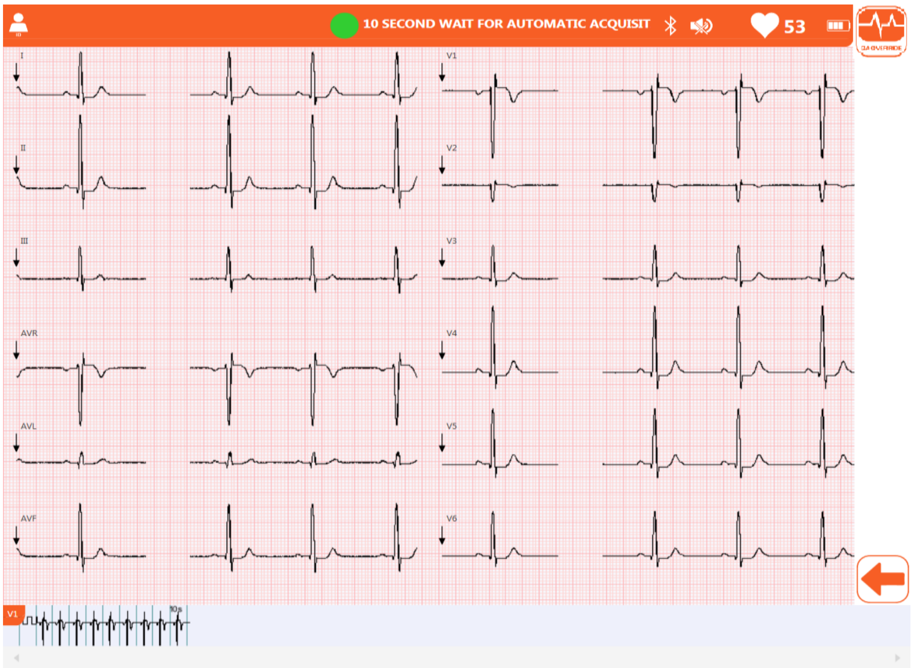
6.6.1. Automatically acquiring an ECG examination

10 s of tracing can be acquired in automatic mode by pressing the **Auto** button in the real time display window or by clicking the central button on the HD+ acquisition device.
Once you press **Auto**, the program acquires 10s of data and creates an examination.

Auto button

During acquisition, the display will show “Waiting for 10s automatic acquisition" in the message area.
touchECG acquires and temporarily stores the entire trace received from the time the HD+ unit was switched on and connected. This way, when you press **Auto**, touchECG analyses the last 10s of trace and if the data are valid (without disconnected leads or dropouts) immediately creates the examination without acquiring other data.
On the other hand, if the last 10s of trace are not valid, the program waits for the next 10s to be acquired. In this case the user can force acquisition of the traces by pressing **QA Override** (this button is enabled once **Auto** has been pressed). This way the electrocardiograph will create an examination of the data acquired up to that moment.
Likewise, if 10s of signal have not yet been acquired, touchECG waits for them to arrive, but in this case you cannot force acquisition (**QA Override** is disabled).

QA Override button



QA Override button enabled

Once the acquisition is completed, the trace is displayed as a printout preview which you can print out, save, edit by changing the patient details, etc., as described in para. 6.7.

Note: The temporary storage capacity is 30 minutes. Once exhausted, touchECG deletes the first minute and continues storing new data. This means that the device's memory always holds the last 30 minutes of signal.

Note: the QA Override button is only enabled after you press Auto, and only if the last 10s of trace are not valid for creating an examination.

6.6.2. Review mode acquisition of an ECG examination (Windows version only)

Click on **Review** in the real time display window to access ECG examination Review mode.
This mode displays a preview of the entire trace recorded since the unit was launched and connected to an HD+ unit up to when you press the **Review** button, for 30 minutes at most.
This preview then allows you to select 10s portions for creating ECG examinations.

Review button

The centre of the window displays a portion of the trace in the selected format, the duration of which depends on the size of the screen. The bottom bar displays the continuous trace. The portion of trace displayed at the centre is highlighted in orange on the continuous trace as shown in figure (**A**).

To change the part of trace displayed, click on the continuous trace (with the right mouse button or double-clicking in the case of a touch screen): this will display a part of the trace centred on the on the point in which you have clicked.
If the recorded trace is long and cannot be displayed entirely in the bottom bar, a scrolling bar appears enabling you to scroll and display the continuous trace.
Bookmarks (par. 6.4.3) are indicated as orange vertical lines.

You can set the speed, amplitude, format and muscle filter applied to the traces by clicking on the respective buttons.
Each button cycles through its available values, and applies them to the traces. For example, the speed button can take the values 5, 10, 25 and 50 mm/s, and each time you click the button it will move to the next value in the sequence.
The button itself displays the value.

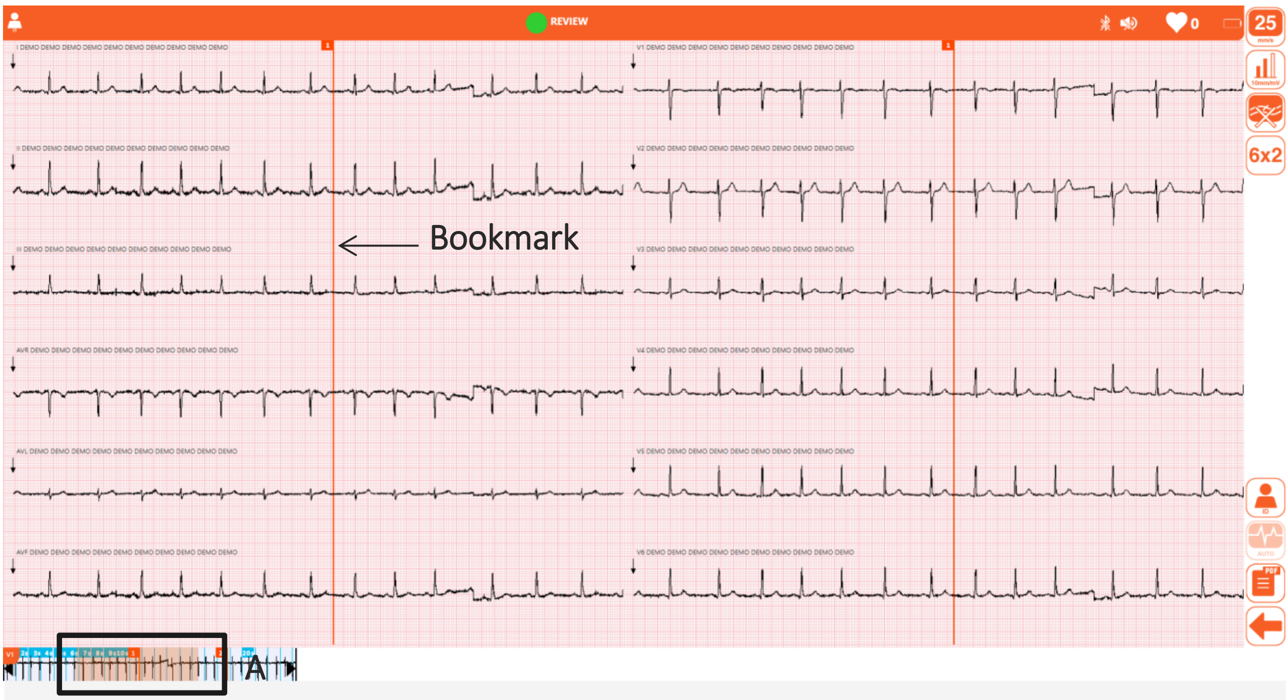
Click on **Id** to set the patient details for the examination, as described in par. 6.5.

10s ECG printout:

Click on the trace in central area to select 10s of trace starting from the point on which you clicked. The 10s portion is highlighted in grey, in both the central area and on the continuous trace.
The **Auto** button is enabled and you can therefore record an examination using the selected 10s portion (par. 6.7).

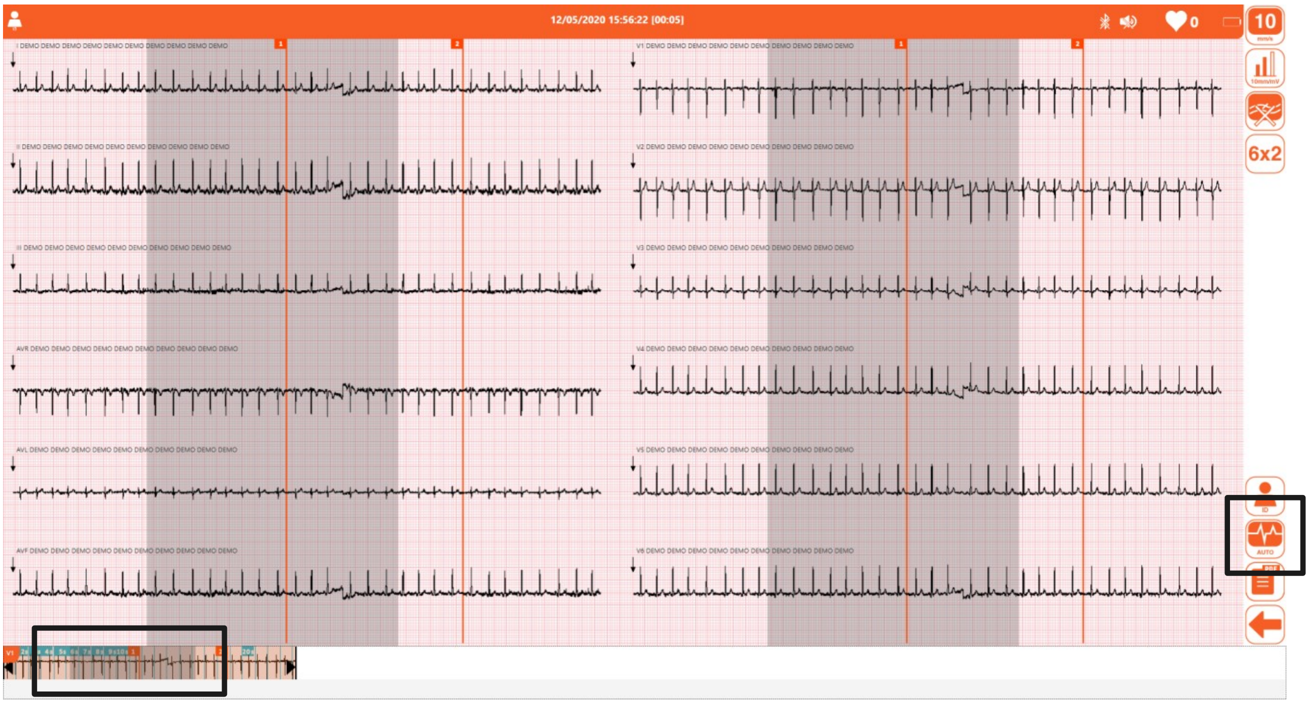
PDF printing of variable duration ECG:

It is possible to save all the recorded track or a part of it by moving the markers on the continuous track and pressing the **Save PDF** button. The save format will respect the following views: 12x1, 3x3, 1x1, while gain, speed and muscle filter will be applied as selected by the user.

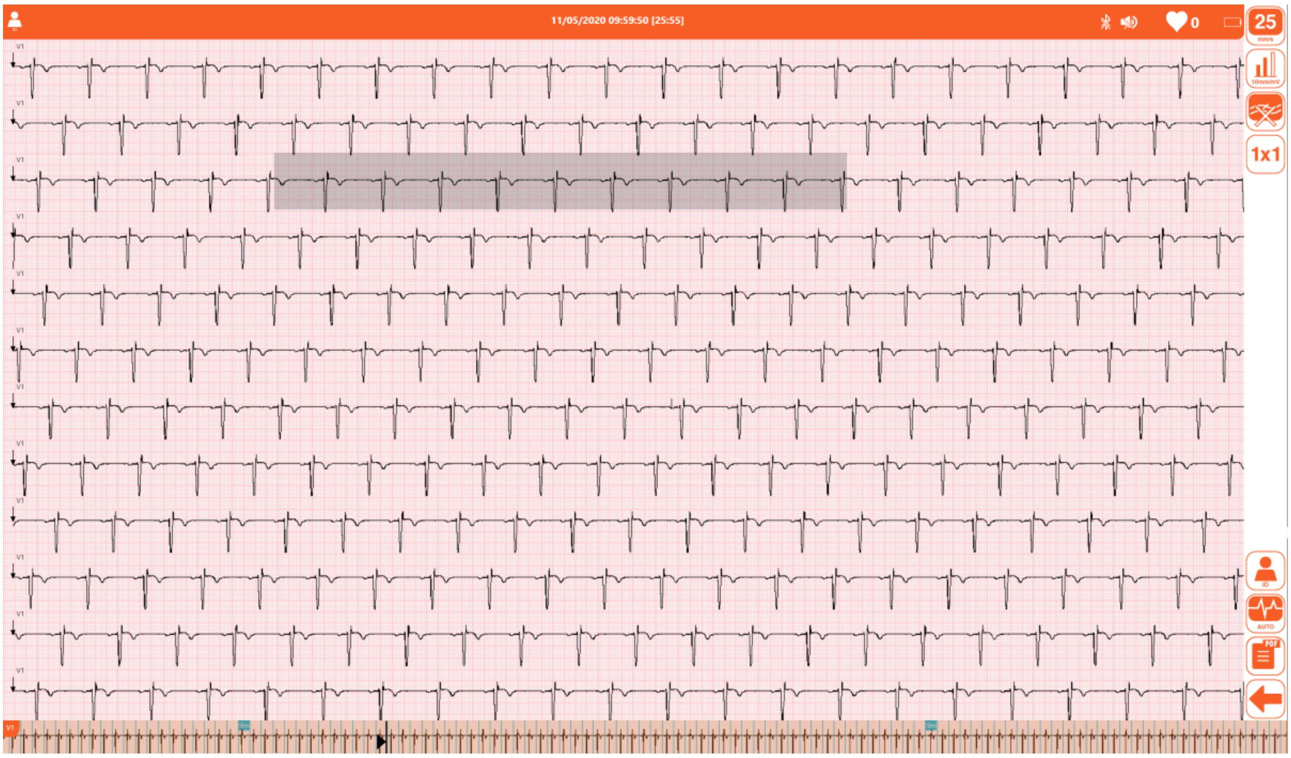


Display in Review mode

CARDIOLINE



Selecting 10s of trace



Saving registration

Buttons – side bar:

- 

Speed — Selects the trace speed
5, 10, 25, 50 mm/s
- 

Amplitude — Selects the trace amplitude
5, 10, 20 mm/mV
- 

Muscle filter — Selects the muscle filter
25 Hz, 40 Hz, 150 Hz, off

  ***Note**: The 150 Hz filter only works on 1000 Hz acquisitions and is only available in the **Windows** version.*

  ***Note**: the 25 Hz filter is stronger than the 40 Hz filter. Setting the filter to "off" disables muscle filtering of the traces.*
  The 150 Hz filter is only applied on 1000 Hz acquisition
- 

Format — 12 leads: 6+6, 3x1 (Ryhtm lead), 12x1, 1x1 (compacted)

  15 leads: 6+6, 3x1 (Rythm lead), 15x1, 1x1 (compacted)
- 

Id — Opens the patient window for entering the patient details (para. 6.5)
- 

Print PDF — To save an exam in PDF format corresponding to the selected path or to the whole exam (if no area has been selected).
- 

Auto — Creates an ECG examination using the selected 10s portion (par.6.7)
- 
Back — Closes the window and returns to the main window in real time with the same data as the patient's personal data or with clean personal data, depending on the user's response to the question proposed.

Tracings display format:

Speed

- 

5 mm/s
- 

10 mm/s
- 

25 mm/s
- 

50 mm/s

Amplitude

- 

5 mm/mV
- 

10 mm/mV
- 

20 mm/mV

Muscle filter

- 

25 Hz
- 

40 Hz
- 

150 Hz — *Note: The 150 Hz filter only works on 1000 Hz acquisitions and is only available in the **Windows** version.*
- 
off

Note: the 25 Hz filter is stronger than the 40 Hz filter. Setting the filter to "off" disables muscle filtering of the traces.

Format

- 

6+6
- 

3x1
- 

12x1
- 

15x1 — Only with HD+ 15 and CLICKECG-HD 15

Note: The temporary storage capacity is 30 minutes. Once exhausted, touchECG deletes the first minute and continues storing new data. This means that the device's memory always holds the last 30 minutes of signal.

6.6.3. Manual mode acquisition of an ECG examination

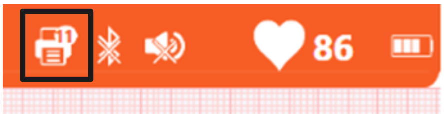
Pressing the **Manual** key on the real time display screen starts continuous printing of the ECG tracing. The printout can be interrupted by pressing **Stop**, which is enabled as soon as manual mode is launched.

The output of the manual print is set from the "Output" field on the Manual Settings tab. It may be, for Windows environments printer, PDF or both, for the Android version only PDF will be possible.

PDF printing will only contain the default rhythm derivation, as well as printing the compacted 1x1.

For Windows versions only:

The remaining number of pages for printing is shown in the top bar with the printer icon. The icon disappears when all of the pages are printed. On each printed page there is also the relative page number.



Pages to print

If the support device's default printer is a PDF file writer, manual printing creates multiple PDF files in sequence.

For both Windows and Android versions:

When the **Manual** print button is pressed, the print process is started or stopped.

Each of the printed pages features:

- Header:
    - On the left: surname, name, ID;
    - In the centre: date and time of acquisition, page number;
    - On the right: cardiac frequency;
- Footer:
    - On the left: amplitude, speed, filters;
    - In the centre: department name
    - On the right: device model, software version.

Buttons – side bar:

- 

  Stop — Stops the manual ECG printout.

6.7. Previewing an ECG examination

After the ECG examination has been created, as described in par. 6.6.1 and 6.6.2, this is shown as a print preview. Bookmarks (par.6.4.3) are indicated as orange vertical lines on the traces.

The print preview shows at the top the boxes with patient's data (**A**), automatic measurements and interpretation (**B**), if enabled on the device.
In the *Android* version it is possible to click on the icon at the top right to collapse or expand this section, so as to leave more room to the tracings.

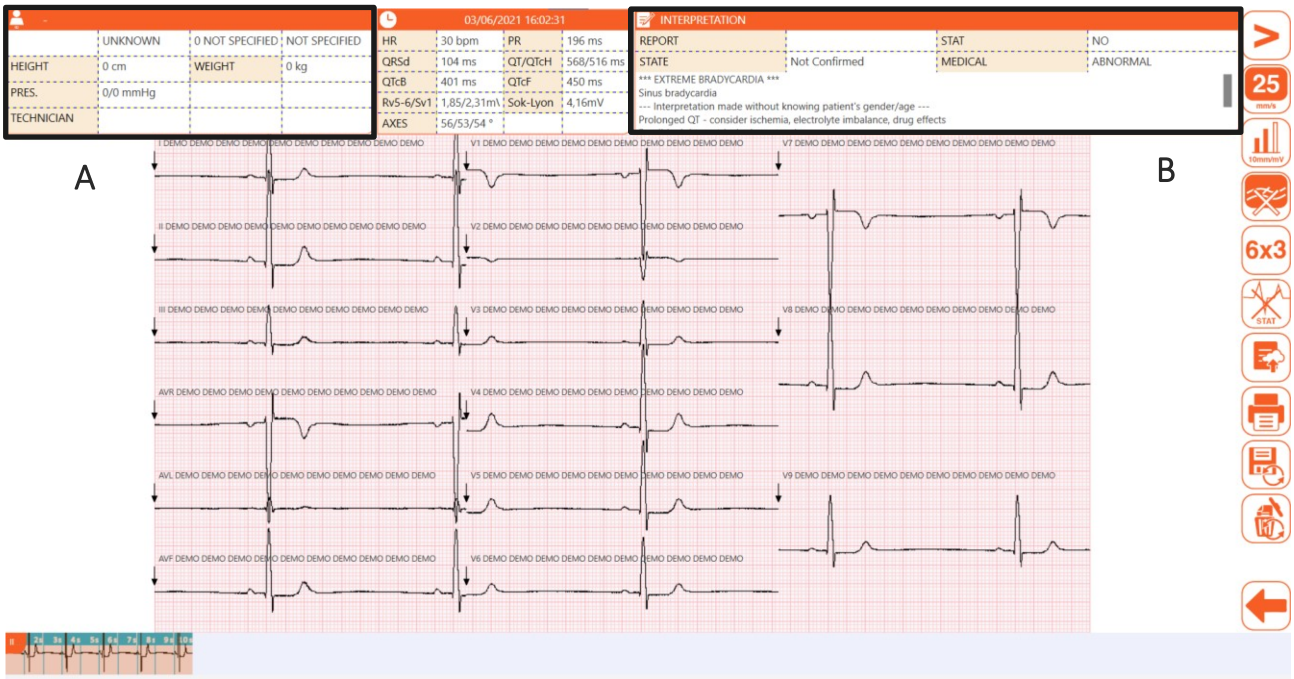
It is possible to change the patient details (clicking on area **A**, see par.6.7.2) and the interpretation (clicking on area **B**, see par.6.7.3), using the calipers tools (enabled/disabled with the **Caliper** button) and zoom (by “pinching” the touchscreen or using the mouse scroll button for a normal screen) as support for the medical report.

*Note: the calipers instrument is only available in the **Windows** version.*

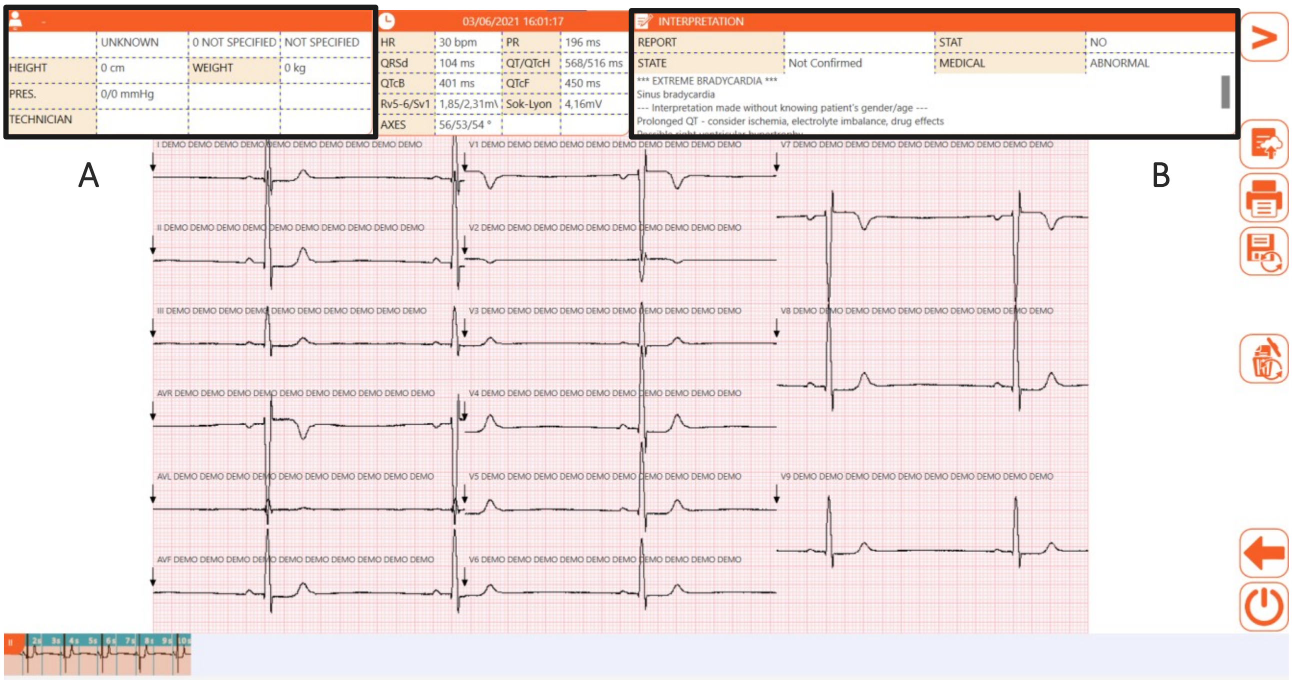
Note: if the acquired signal were to show consecutive signal losses of at least 100 ms or an overall loss of 1 s, the test would be classified as low quality and this indication would appear in the interpretation box.

If full user interface mode is enabled (Quick user interface setting = Off) it is also possible to change the speed, amplitude, format and muscle filter applied to the tracings by clicking on the corresponding buttons (see par. 6.7.1). The changes are applied to both the preview and the printed and saved examination (if any).

The **Urgent** button enables you to attribute the “Urgent” status to the examination, as described in par. 6.7.4, while the **Transmit**, **Print** and **Save** buttons enable you to transmit the examination to an external system, print and save it (par. 6.7.4 and 6.7.6).
Finally, the **Launch App** button launches an external application, as configured in the Settings (e.g. a web browser), while **Send email** sends the examination by email as a PDF (par. 6.7.7).



Examination preview window with Full user interface



The preview window with Quick user interface

Available buttons – side bar:

C indicates the controls only featured in the full user interface
W indicates the controls only featured in the *Windows* version

Menu 1

-  Open menu — Opens secondary menu, Menu 2.
- 

C Speed — Selects the trace speed
25, 50 mm/s
- 

C Amplitude — Selects the trace amplitude
5, 10, 20 mm/mV
- 

C Muscle filter — Selects the muscle filter
25 Hz, 40 Hz, 150 Hz, off
Note*: The 150 Hz filter only works on 1000 Hz acquisitions and is only available in the* ***Windows*** *version.*
Note*: The 25 Hz filter is stronger than the 40 Hz filter. Setting the filter to "off" disables muscle filtering of the traces.*
- 

C Format — Selects the trace format
12 leads: 6x2, 3x4, 3x4+1, 3x4+3, 12x1
15 leads: 6x3, 3x5, 3x5+1, 3x5+3, 15x1
- 

C Urgent — Turns the "urgent" status of the examination on/off.

▪		Transmit	Transmits the acquired examination to an external system
▪		Printing	Prints out the acquired examination
▪		Save / Update	Saves / updates the acquired examination
▪		Erase and repeat	It allows you to delete the exam from the local archive and keep the same personal data of the patient by returning to real-time mode.
▪ 		Back	Closes the window and returns to the main window in real time with the same data as the patient's personal data or with clean personal data, depending on the user's response to the question proposed.

Menu 2

▪		Close menu	Closes the secondary menu and returns to Menu 1
▪	W	Calipers	Enables/disables the calipers tool *Available in the* **Windows** *version only.*
▪		Send email	Opens a new email message to send the acquired examination as a pdf
▪ 	W	Launch App	Launches an external application (if so configured in the settings). *Available in the* **Windows** *version only.*

6.7.1. Changing the display and printout mode

If full user interface mode is enabled (“Quick user interface” setting = Off) it is possible to change the speed, amplitude, format and muscle filter applied to the tracings by clicking on the corresponding button. The display mode applies to both the printout and the saved file (if any).
Each button cycles through its available values, and applies them to the traces. For example, the speed button can take the values 25 and 50 mm/s, and each time you click the button it will move to the next value in the sequence.
The button itself displays the value.

All possible values of the parameters and the button labels are listed below:

Tracings display format:

Speed

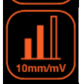
- 

50 mm/s
- 

25mm/s

Amplitude

- 

5 mm/mV
- 

10 mm/mV
- 

20 mm/mV

Muscle filter

- 

25 Hz
- 

40 Hz
- 

150 Hz

*Note: The 150 Hz filter only works on 1000 Hz acquisitions and is only available in the **Windows** version.*

- 
off

Note: the 25 Hz filter is stronger than the 40 Hz filter. Setting the filter to "off" disables muscle filtering of the traces.

Format

- 

6x2 — 5 seconds of 6 leads in 6 channel format.
- 

3x4 — 2.5 seconds of 12 leads in 3 channel format.
- 

3x4+1 — 2.5 seconds of 12 leads in 3 channel format; the fourth channel is a rhythm strip of 10 second lead defined by the user (Rhythm Traces).
- 

3x4+3 — 2.5 seconds of 12 leads in 3-channel format plus a strip of 10 second leads defined by the user (Rhythm traces), in a 3-channel format.
- 

3x5 — 2 seconds of 15 leads in 3-channel format
- 

3x5+1 — 2 seconds of 15 leads in 3-channel format; the fourth channel is a 10-second rhythm strip of lead defined by the user

(Rhythm leads).

- 3x5 +3 — 3x5+3 — 2 seconds of 15 leads in 3-channel format plus a 10-second strip of user-defined leads (Rhythm leads), in a 3-channel format.
- 12x1 — 12x1 — 10 seconds of 12 leads in 12 channel format.
- 15x1 — 15x1 — 10 seconds of 15 leads in 15 channel format. Only with HD+ 15 and CLICKECG-HD 15.
- AVG — AVG — Medium complexes

6.7.2. Modifying patient details

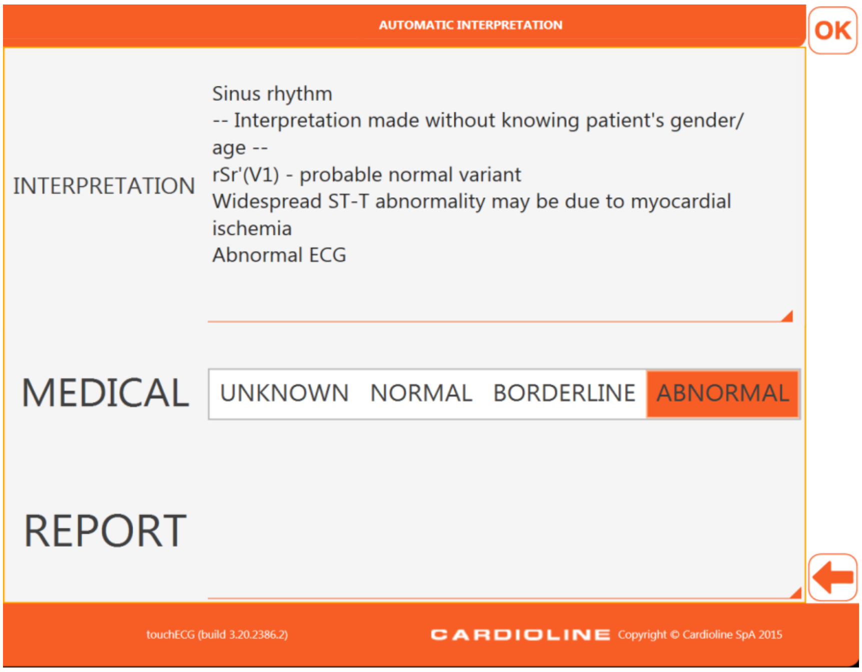
You can open the Patient window to modify the current patient data by clicking on the patient **ID** area as show in figure (**A**).
The data can be entered/edited manually as described in par.6.5.2 or from a work list, as described in par. 6.5.4.

Note*: if you access the Patient window from the examination preview window, the automatic patient details entry options from the examinations archive are not available.*

6.7.3. Modifying the automatic interpretation

If the program's interpretation option is enabled and “Automatic interpretation” is enabled in Settings, an automatic interpretation is added to the examination. With the “Automatic interpretation” function (Par. 8.2.2) you can choose whether to display the automatic interpretation text in summary or extended mode, or not display it at all.
The interpretation can be modified, or entered if not present, by clicking on the **Interpretation** area, as shown in figure (**B**).
The automatic interpretation window allows you to edit the interpretation text, assign an overall evaluation to the examination (Unknown, Normal, Abnormal, Borderline) and enter the name of the doctor.

The acquired examination is in the “Not Confirmed” status until the name of the doctor is entered in the “Doctor” field of the Automatic Interpretation window. This way the examination switches to the “Confirmed” status. Vice-versa, if the “Doctor” field remains empty or is deleted, the examination switches to the “Not Confirmed” status.



Window for editing the interpretation

Buttons

-  OK — Closes the window, saves the data and changes the test status to “Confirmed” if the “Doctor” field is filled in. The current printing format (speed, amplitude, sampling rate, trace format, muscle filter) is also saved in the test.

  If the “Doctor” field is empty, the test will be “Not confirmed” and any previously saved printing format will be deleted.
-  Back — Closes the window without saving the data and leaves the examination's status set to “Not Confirmed".

Note: if the name of a doctor is not entered, the status of the examination remains “Not Confirmed”.

6.7.4. Identifying an urgent ECG examination

The acquired examinations can be identified as “urgent”, for example for flagging as such in Cardioline ECGWebApp or in an external archive.

Once the examination has been created, you can flag it as “urgent” by clicking on **Urgent**. When this key is pressed, the corresponding icon changes and the acquired examination is given the "urgent" status. Pressing the key again toggles the "urgent" status.

***Urgent** on/off button*

If the automatic saving option is enabled, the examination is re-saved automatically every time the "urgent" status is modified (i.e. every time the **Urgent** button is pressed). If on the other hand the automatic saving option is not enabled, the examination must be saved manually at each modification by pressing the **Save/Update** button.

If an examination must be acquired urgently, it can be started without entering the patient's data, but simply by pressing the **Auto** key on the real time display screen.
The resulting examination can be printed, saved or transmitted as it is without the patient's details, or you can enter the details as described in para. 6.7.2.

You can enable the “Auto Stat” option in Settings to flag all acquired examinations as urgent. This is useful when the program is used in an ambulance or an emergency ward. You can always cancel the urgent flag by pressing the **Urgent** button.

Note*: To transmit the examination as urgent, it must be sent after having modified the status.*

6.7.5. Printing out and saving an ECG examination

If the automatic printing option is enabled (par. 8.2), the examination is printed automatically at the end of the acquisition. Likewise, if the automatic saving option is enabled (par. 8.2), the examination is saved automatically at the end of the acquisition.
If these options are not enabled, you can print out and save the examination with the **Print** and **Save/Update** buttons (to save an examination or overwrite an existing one).

If you press **Save/Update**, the examination is saved in the local touchECG database and, if the settings are correct (par. 8.2), on the computer as well, as an SCP file and/or PDF in the specified folders.

Each of the printed pages features:

- Header:
    - On the left:
        - patient data: ID, surname, name, gender, date of birth, race;
        - examination data: height, weight, age, systolic and diastolic blood pressure, SpO2, medications, notes, technician, order number
    - In the centre: automatic measurements;
    - On the right: automatic interpretation (if enabled);

- Footer:
    - On the left: amplitude, speed, filters;
    - In the centre: department name
    - On the right: device model, software version.

Note: if the support device's default printer is a PDF file writer, printing launches this writer which creates a PDF file.

Attention: If you are printing PDF files, you need to set up a corresponding program so that the document is in no way adapted or scaled. If you are using Acrobat Reader, you need to select "Actual Size" in "Page management and size." Otherwise you may get non diagnostic quality prints.

Note: In case of Android operating systems, you should configure the printer first as described in section Errore. L'origine riferimento non è stata trovata..

6.7.6. Transmitting an ECG examination

touchECG can transmit an examination to an external system (e.g. a Cardioline ECGWebApp system, an examination management system, a CIS/PCS DICOM system, etc.), as described in par. 7.

If the automatic transmission option is enabled (par. 8.2), the examination is transmitted automatically at the end of the acquisition. If the program fails to transmit the examination (no network connection, for instance), the automatic transmit function periodically attempts to send the untransmitted examinations until it is successful in doing so.

If the automatic transmission option is not enabled, you can transmit the examination by pressing the **Transmit** button. See par. 7.3 for further details.
If one or more tests fail/s to send the application will try sending again until it is successful. When the examination has been transmitted, the system displays the message "Examination transmitted".
If the user tries to close the application when there are still tests that need to be sent, a message window will open warning the user that there are unsent tests.

Icon for tests to be sent

You can also transmit a specific examination or all untransmitted examinations from the examinations archive window, as described in par.6.8.

Note: if the automatic transmit function is enabled, the automatic save function is also enabled.

Note: if the delete test after sending feature is enabled (only available if the Privacy option is active) the test(s) is deleted after being transmitted successfully.

6.7.7. Sending an ECG examination by email

At the end of acquisition, you can send the examination by email by clicking on **Email** in Menu 2.
touchECG creates and opens a new email message in the computer's default email application, and attaches the examination as a PDF report.

You can also send the examination by email from the examinations archive window, as explained in par. 6.8.

***Note**: in the **Windows** version, to send the examination by email, your email application must support the EML format.*

***Note**: in the **Windows** version, if no email application is configured, you will be prompted to open the "eml" file with an application.*

6.8. Test Archive

touchECG has an internal archive of completed examinations, which can store up to 1000 ECG's.
The examinations can be saved automatically or manually at the end of acquisition, as described in par. 6.7.4
To access the archive and display the list of saved examinations, open the Examinations Archive window by clicking **Examinations Archive** in Menu 2 of the real time display window.

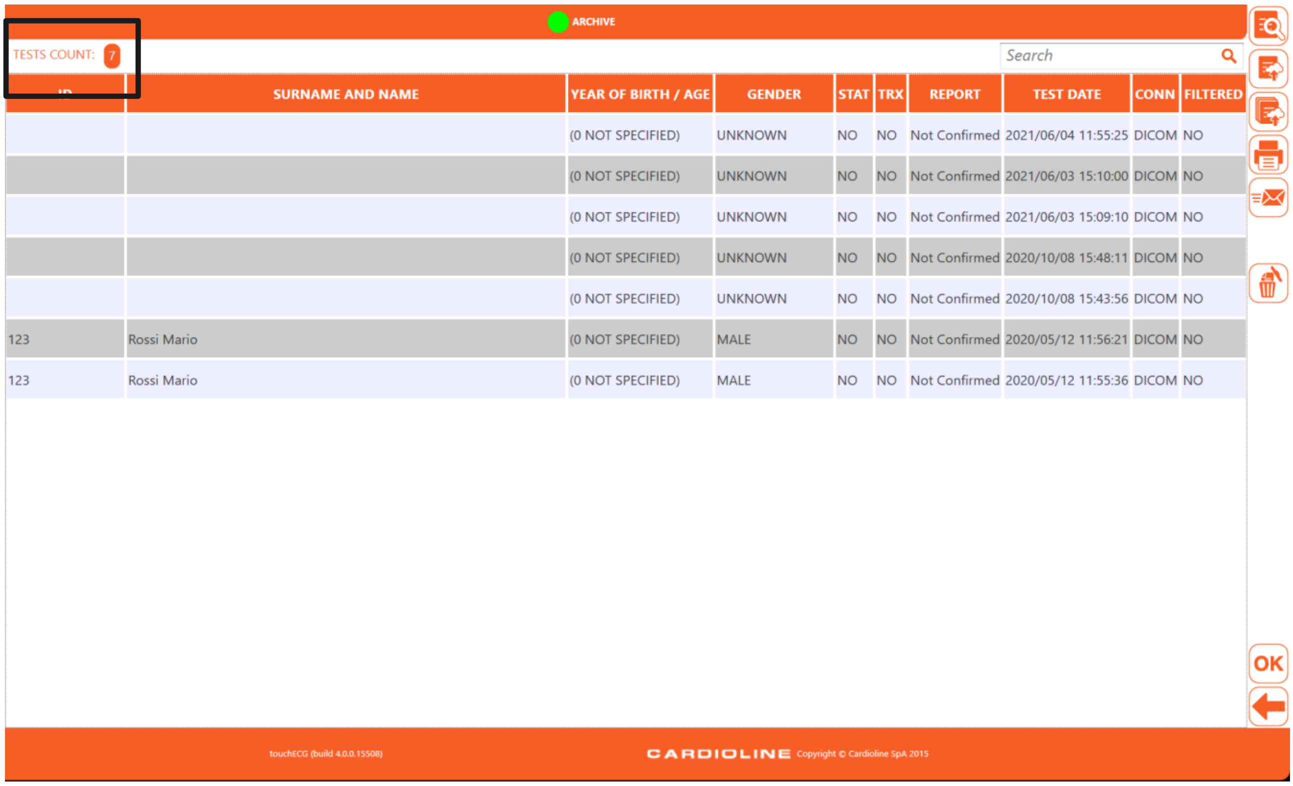
Opening the Examinations Archive window

The top left of the screen displays the number of examinations in the archive.
The following fields are displayed in order for each examination:

- Id: the patient's identification code
- Name and surname: the patient's name and surname
- Year of birth / age: the patient's year of birth and age
- Gender: the patient's gender
- Stat: the examination's urgent flag (Yes = urgent examination)
- Trx: the examination's transmitted flag (Yes = examination transmitted)

- Report: indicates whether the interpretation of the examination has been confirmed by a doctor or not
- Date of examination: date and time of the examination
- Connectivity: connectivity option associated with the examination (linked to the HD+ unit used to acquire the ECG, as described in par. 10.2).
- Filtered: indicates whether the exam is filtered, that is, if it contains the waveforms filtered with the muscle filter. If this is the case, no LP (muscle) filter can be applied anymore while viewing the exam.

Click on the column labels (e.g. “ID” or “Name and Surname) to order the list in increasing/decreasing order by the field in question.



Examinations archive window

The “Search" field allows you to search for an examination containing the string entered in the “ID” or “Name and Surname" fields.
In the Test Archive window you can select a test by clicking on the corresponding line to display it, transmit it to an external system, print it out, send it by email as a PDF report or delete it, using the respective buttons.
You can also select multiple tests (only for Windows version) by clicking on the corresponding lines while holding down the CTRL key. It is therefore possible to perform the same action on all selected tests using the corresponding buttons: transmit them to an external system, print them out or delete them.

If you click on **Delete** you are prompted to confirm the action.
Displaying the examination opens the preview window described in par. 6.7.

Transmit all transmits all so far untransmitted examinations at the same time.
If you select an examination, you can automatically load the patient details associated with it by pressing **Ok**. This closes the Examinations Archive window and loads the patient details into the real time display window.

Buttons

Display	Opens and displays the selected examination in preview mode (par. 6.7)
Transmit	Transmits the selected test to an external system (see par. 7.3) If the "Delete after sending" setting is enabled, the test is deleted after transmission.
Transmit all	Transmits all non-transmitted tests to an external system (par. 7.3) If the "Delete after sending" setting is enabled, the tests are deleted after transmission.
Printing	Prints out the selected examination
Send email	Sends the selected examination by email (as a PDF report)
Remove	Deletes the selected examination
OK	Closes the window and loads the patent data associated with the selected examination.
Back	Closes the window without loading the patent data associated with the selected examination.

7. CONNECTIVITY, RECEIVING WORK LISTS AND TRANSMITTING ECG EXAMINATIONS

7.1. General information

Using the support device technology (LAN, WiFi, GPRS, etc.), touchECG can connect to the internet and external systems to receive work lists and transmit acquired examinations.

As described in par. 4.4, touchECG may be equipped with two different connectivity options, with different protocols:

- **Standard connectivity**: for connection to external systems using a standard internet protocol (http/https). In this mode, touchECG can connect to Cardioline ECGWebApp or other on-line systems (e.g. GDT systems).
  The following protocols are available:
  - Receiving work lists:
    - Cardioline: for connection with Cardioline ECGWebApp or with other systems in the network
    - GDT: for connection to GDT systems (only available in the **Windows** version)
    - Text file: for importing work lists generated by other systems (only available in the **Windows** version)
    - XML: to connect with systems that support this Protocol (only available in the **Windows** version)
  - Transmitting examinations
    - Cardioline: for connection with Cardioline ECGWebApp or with other systems in the network
    - GDT (Standard): for connection to GDT systems (only available in the **Windows** version)
    - XML: to connect with systems that support this Protocol (only available in the **Windows** version)

- **DICOM connectivity** (optional): for connecting to external systems running the DICOM protocol. This allows you to integrate the system into any PACS DICOM system for managing hospital work flows.
  The following protocols are available:
  - Receiving work lists:
    - Cardioline: for connecting with DICOM systems

- Transmitting examinations
  - Cardioline: for connection with Cardioline ECGWebApp

It is possible to implement other communication protocols, as well as allow the Android version to communicate through the GDT and XML protocols.
Through ad hoc projects it is also possible to include other communication formats and protocols.

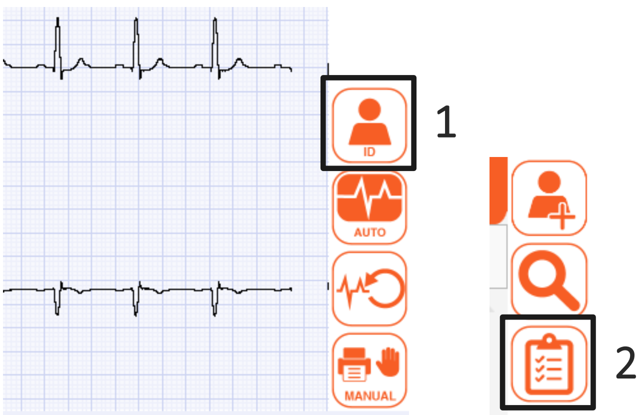
touchECG, finally, can export examinations as SCP and PDF files to specific folders on the computer and send PDF examination reports via email.

***Note**: as indicated in par. 10, the connectivity options (Standard / DICOM) as well as the interpretation options (none / Glasgow) are not linked to the touchECG software but to the HD+ acquisition unit in question. It follows that whether an examination is transmitted in Standard or DICOM mode depends on the HD+ unit used to acquire the examination.*

7.2. Receiving a work list

touchECG can receive work lists from external systems, from which you can select the order corresponding to the examination to be administered and thus automatically load the patient details associated with the examination.

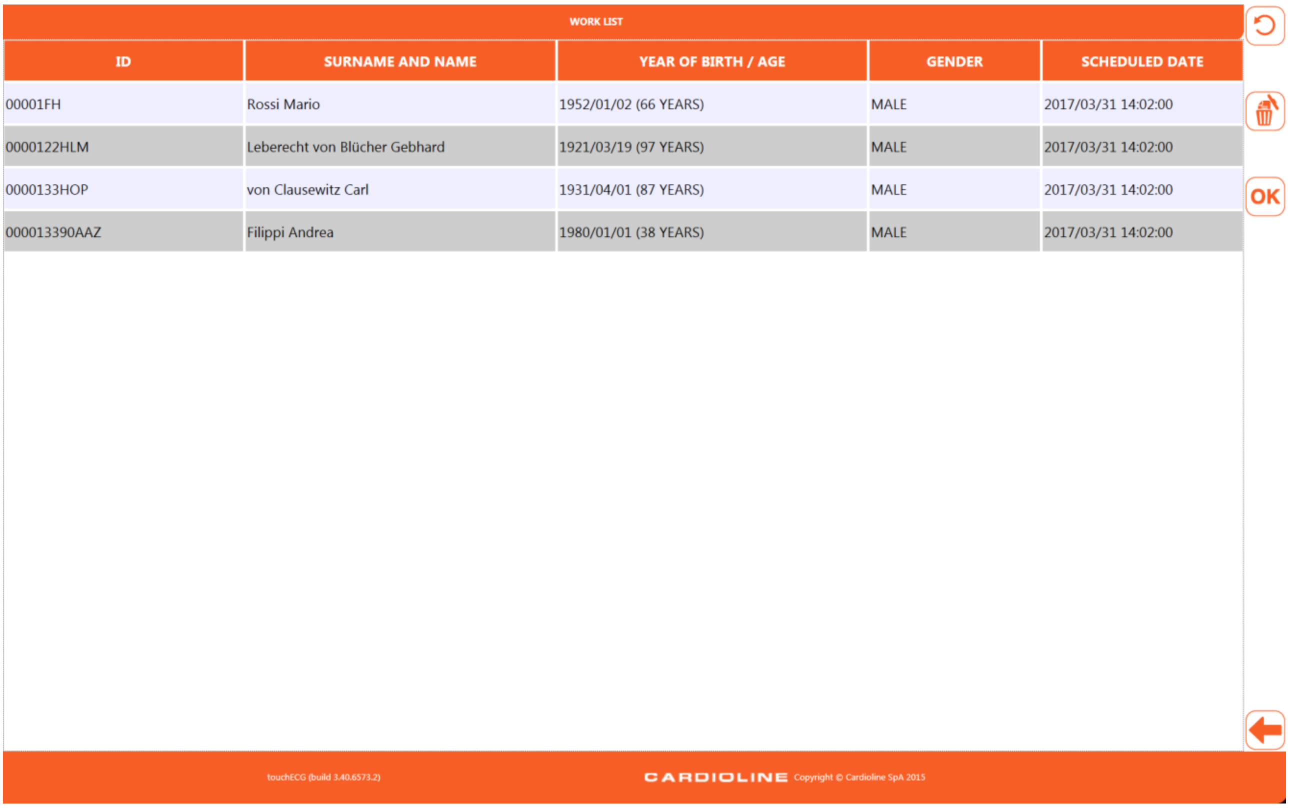
To receive a work list, click on **Id** in the real time display window (**1**) and then **Work List** in the Patient window (**2**), as described in par. 6.5.



***Id** button in the real time display window and **Work list** button in the Patient window.*

The work list is downloaded and displayed, and you can select an order to load (thus also loading the patient details) or delete from the list.
See par. 6.5.4 for further details about work lists.

The work list can be acquired automatically (option -g) using the command line and, if necessary, always automatically by launching the acquisition of a test in **Auto** mode (option -a) ; while if you only want to go directly to the work list screen without making the choice of the first patient, you can use the -w option.

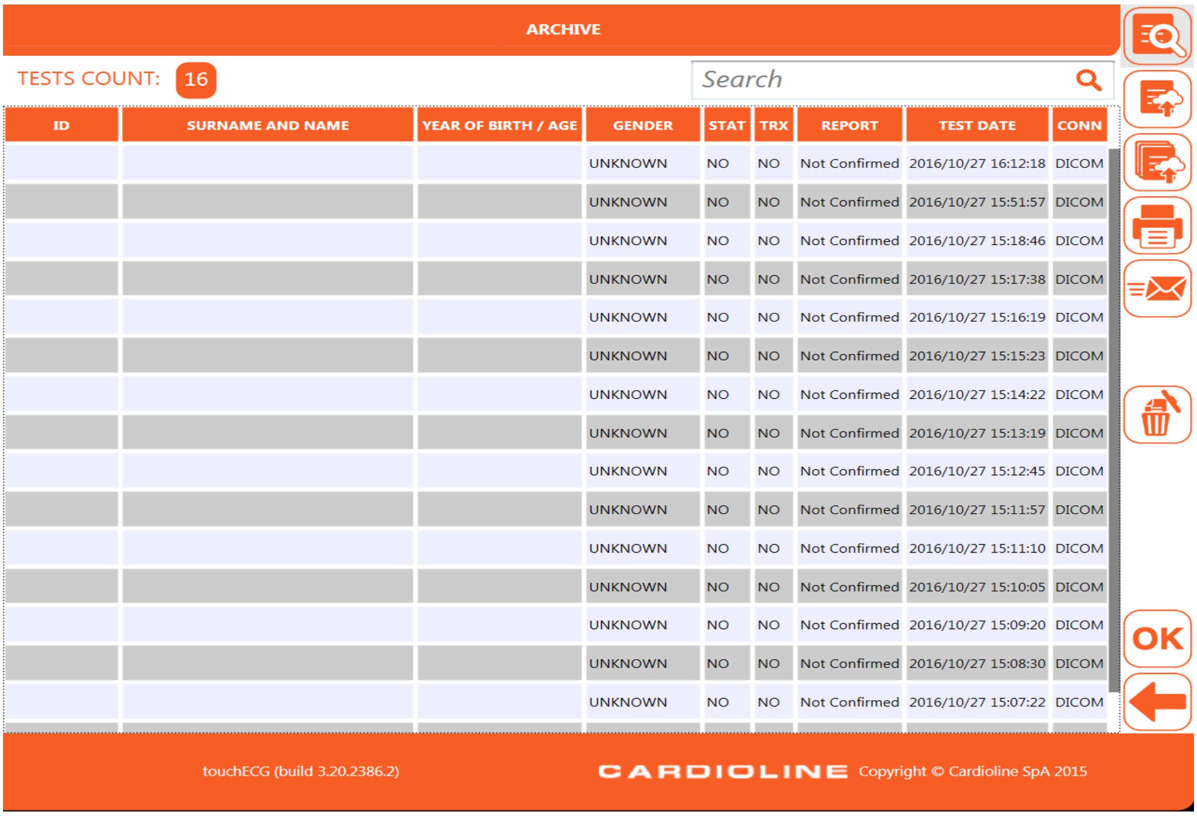

WORK LIST

ID	SURNAME AND NAME	YEAR OF BIRTH / AGE	GENDER	SCHEDULED DATE
00001FH	Rossi Mario	1952/01/02 (66 YEARS)	MALE	2017/03/31 14:02:00
0000122HLM	Leberecht von Blücher Gebhard	1921/03/19 (97 YEARS)	MALE	2017/03/31 14:02:00
0000133HOP	von Clausewitz Carl	1931/04/01 (87 YEARS)	MALE	2017/03/31 14:02:00
000013390AAZ	Filippi Andrea	1980/01/01 (38 YEARS)	MALE	2017/03/31 14:02:00

Work List display window

7.3. Uploading an exam

TouchECG can transmit examinations to an external system via internet.
The examinations can be transmitted automatically or manually at the end of acquisition, as described in par. 6.7.6, or by accessing the Examinations Archive by selecting the examination in question and clicking on Transmit. From the Examinations Archive you can also transmit all untransmitted examinations together with Transmit all.
See par. 6.8 for further details about the Examinations Archive.



Examinations archive window

7.4. Sending an examination by email

Examinations can be emailed as PDF attachments.
This can be done either from the preview window at the end of acquisition (par. 6.7), or from the Examinations Archive (par. 6.8) using the **Send email** button.

***Send email** button*

***Note**: in the **Windows** version, to send the examination by email, your email application must support the EML format.*

***Note**: in the **Windows** version, if no email application is configured, the message "Could not send, no email application installed" will be displayed.*

7.5. Saving an examination in SCP and PDF format

Once an ECG has been acquired, it can be saved automatically or manually, as described in par. 6.7.4.
In particular, at each save (automatic or manual) the examination is also exported as a SCP and/or PDF file in the folder indicated in "scp folder" and "pdf folder" in the OTHER tab in Settings (par. 8.2). If no folder is specified, the examination is saved in the default folder:

Windows:
C:\Users\[*NomeUtente*]\Documents\touchECG\scp
C:\Users\[*NomeUtente*]\Documents\touchECG\pdf

Android:
/storage/emulated/0/Documents/touchECG/scp
/storage/emulated/0/Documents/touchECG/pdf

If you set the "Filename format" you can specify the structure of the filenames of saved PDF and SCP files.

The "Automatic SCP export" and "Automatic PDF export" options automatically export the examinations after they have been acquired.

You can also export the SCP / PDF files after completing the examination, by clicking Display in the Examinations Archive. This opens the examinations Preview window, which contains the commands for processing examinations, including saving (see para. 6.7 and 6.8).
If the setting "Include filtered data in the SCP file" is enabled (see par. 8.2.6) the filtered tracks are saved in the SCP file, with the same filter applied on the screen. Vice versa, the raw data are saved.

7.6. Connectivity formats and protocols

7.6.1. GDT (in the *Windows* version only)

GDT mode allows you to connect to systems using the GDT format.
You can then receive worklists and send examinations in GDT format.

To launch TouchECG automatically from a GDT system, launch the program from the command line as described in Par. 5.6.

To connect to GDT systems, click on "GDT" in the "Receiving protocol", "Transmission protocol" and "Protocol configuration" fields in the CONNECTIVITY tab in Settings (see para. 8.2)
You must also configure the fields which appear after having selected "Protocol Configuration":

- Work list folder: path of the folder containing the work lists created by the GDT system
- Examinations folder: path of the folder in which touchECG saves the GDT format examinations, for uploading to the GDT system

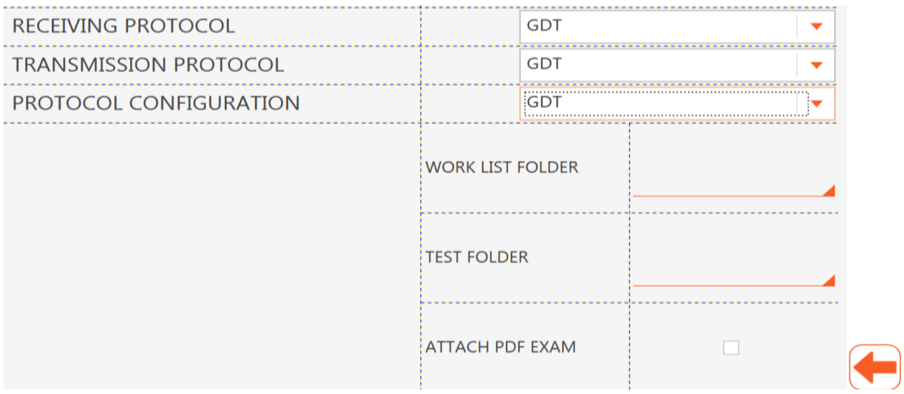
ATTENTION: if the name of the file with the GDT extension is also entered in the path of the exam folder, the system creates a file with exactly that name while otherwise a causal name with the GDT extension is created.

Eg :

Exam folder C: \ GDT \ OUT \ RX_EDP.GDT => a file is created in the chosen folder with the name RX_EDP.GDT

Exam folder C: \ GDT \ OUT => a file is created in the chosen folder with the causal name ADNAN £ 432423DFDSF £ 43.GDT

- **Attach PDF examination**: if enabled, allows you to attach a PDF examination report to the GDT examination file



GDT settings

As mentioned, the structure and content of the PDF report which is attached to the GDT test is defined according to the touchECG settings.

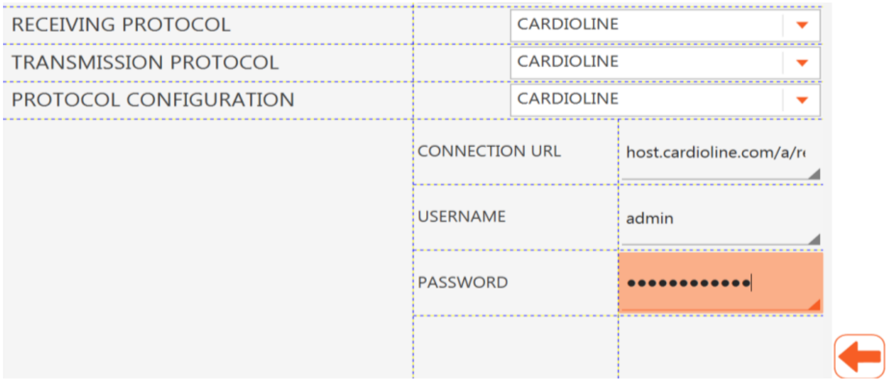
7.6.2. Cardioline Standard

If the Standard Connectivity option is available, Cardioline mode allows you to connect to the Cardioline ECGWebApp application.
This means you can receive work lists prepared in Cardioline ECGWebApp and send SCP format examinations to it.

To connect to Cardioline ECGWebApp, click on “Cardioline” in the “Receiving protocol”, “Transmission protocol" and “Protocol configuration" fields in the CONNECTIVITY tab in Settings (see para. 8.2)
You must also configure the fields which appear after having selected “Protocol Configuration”:

- **Connection URL**: address of the server running Cardioline ECGWebApp to which you wish to connect
- **Username**: username used by touchECG to automatically access Cardioline ECGWebApp (this must be a user authorised to transmit examinations – for details, refer to the Cardioline ECGWebApp user manual)

- **Password**: password for the username with which touchECG automatically accesses Cardioline ECGWebApp



Cardioline connectivity settings

7.6.3. Cardioline DICOM

If the DICOM connectivity option is available, touchECG can connect to external DICOM systems.

To connect in DICOM mode, click on “Cardioline” in the “Receiving protocol”, “Transmission protocol" and “Protocol configuration" fields in the CONNECTIVITY tab in Settings (par. 8.2)
You must also configure the fields which display when you click on “Protocol configuration":

- **Connection URL**: IP address
- **Username**: not required
- **Password**: not required

7.6.4. Text file (in the *Windows* version only)

Text file mode allows you to receive work lists as text files.
For each order in the work list there must be a text file containing the patient details with a structure similar to the following:

```
ID        FIRSTNAME   LASTNAME     SEX          BIRTHDATE    RACE
WEIGHT    WEIGHTUM    HEIGHT       HEIGHTUM     SYSTOLIC     DYSTOLIC
DRUG      NOTES       TECHNICIAN   EMAIL        ADDRESS      ACCESSIONNUMBER
STAT
123456    mario       rossi        119730512    1
90        0           190          0            80           120
drugs     notes       Dr. Bianchi  m@email.it   via linz 2   acess01
1
```

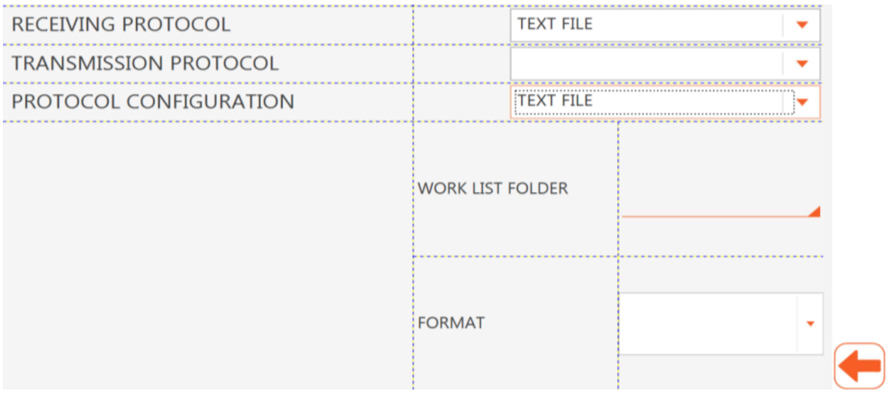
All or only some of the fields may be present, and they must be separated by tab characters. The fields correspond to the following details and take the following values:

- ID: Patient ID – alphanumeric string
- Firstname: First name – alphanumeric string
- Lastname: Last name – alphanumeric string
- Sex: Sex – 0 = unknown, 1 = male, 2 = female
- Birthdate: date of birth – string in format YYYYMMDD (e.g. 19901231: year = 1900, month = 12, day = 31)
- Race: Race – 0 = not specified , 1 = caucasian, 2 = black, 3 = oriental
- Weight: Weight – integer, followed by unit 0 = kg, 1 = lb
- Weightum: Unit of measurement of weight – 0 = kg, 1 = lb
- Height: height – integer
- Heightum: Unit of measurement of height – 0 = cm, 1 = in
- Systolic: Systolic pressure – integer
- Diastolic: Diastolic pressure – integer
- Drug: Pharmacological therapy – alphanumeric string
- Note: Additional remarks – alphanumeric string
- Technician: Name of technician administering the examination – alphanumeric string
- Email: Email address of patient – alphanumeric string
- Address: Physical address of patient – alphanumeric string
- Accession number: Reservation number – alphanumeric string
- Stat: “urgent” examination status – 0 = not urgent, 1 = urgent

To import a work list as a text file, select “Text file” in the “Receiving protocol” and “Protocol configuration” fields in the CONNECTIVITY tab of Settings (par. 8.2)
You must also configure the fields which display when you click on “Format configuration":

- Work list folder: folder containing the text format work list
- Format: indicates the format of the work list (only “Standard” is currently available)

Note: Text file mode is only available for receiving work lists, it cannot be used for transmitting examinations.

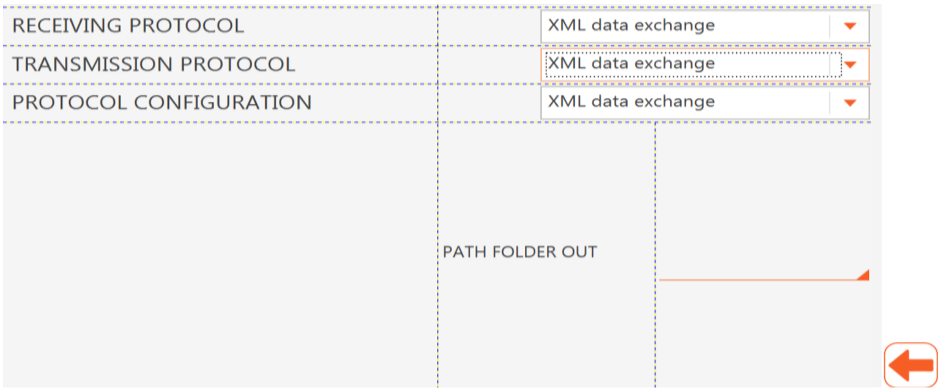


Text file settings

7.6.5. XML (in the *Windows* version only)

XML mode allows you to connect to systems using the XML format.
You can then receive work lists in XML format and send tests in XML format.
To connect in XML mode, select "XML" in the "Receive protocol", "Send protocol" and "Protocol configuration" fields in the CONNECTIVITY tab of the Settings (see par.8.2)
You must also configure the fields which appear after having selected "Protocol Configuration":

- **Examinations folder**: path of the folder in which touchECG saves the XML format examination file.



XML settings

8. DEVICE SETTINGS

8.1. General information

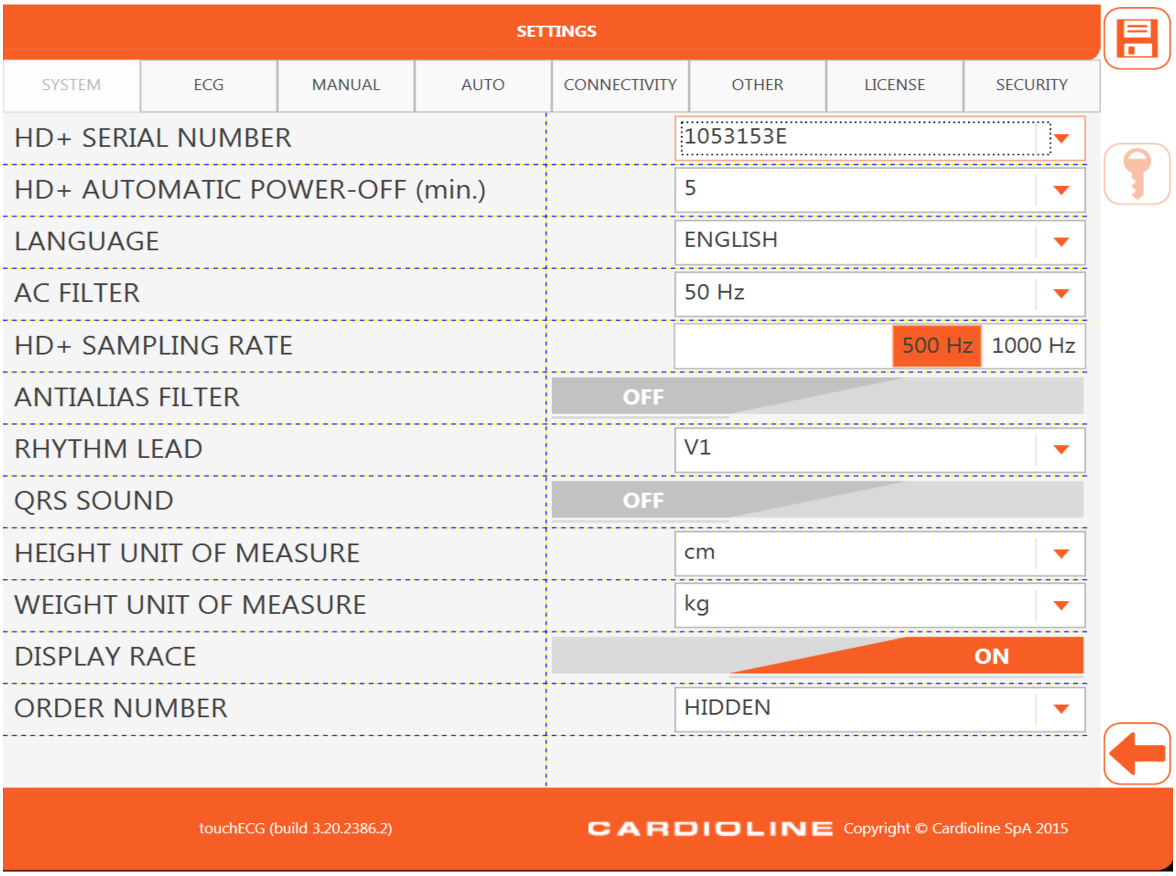
The Settings window is accessible from the real time display window, by pressing **Open menu** (to open the secondary menu) and then **Settings**.

Opening the Settings window

The Settings window has several panels, each of which has data entry fields:

- **System**: basic touchECG settings (HD+ unit, language, AC filter, etc.);
- **ECG**: signal acquisition settings (muscle filter, QTc, type of interpretation, etc.);
- **Manual**: trace display and manual printout settings (grid, trace format, speed, etc.);
- **Auto**: automatic printout settings (trace format, speed, number of copies, etc.);
- **Connection**: connectivity settings;
- **Other**: other settings (save mode, bookmarks, etc.);
- **License**: lists enabled options and updates.
- **Security**: set a security code (PIN) that allows access to the settings section to be protected.

Use **Save** and **Back** to save or cancel any changes.



Settings Window

Buttons

- **Update license** — Updates the program active options (Interpretation / DICOM). Active only in the “License” tab.
- **Save** — Closes the window and saves the changes.
- **Back** — Closes the window without saving the changes.

8.2. Summary of settings

8.2.1. System

Field	Function	Possible values
Connection type	Bluetooth connection type: gives information about the connection type with HD+	Virtual (DEMO), Dongle V1, DongleV2, HD+, HD+12/15, USB
HD+ serial number	Serial number of the HD+ unit you wish to connect	Dropdown editable menu with a list of HD+ acquisition units paired with the computer
HD+ auto off	Time after which, if there is no acquisition, the HD+ acquisition device is switched off automatically.	Deactivated / 5 min / 15 min / 30 min
Language	Language used for display and printing.	English / Italian / German / French / Spanish / Portuguese / Czech / Turkish / Russian / Polish
AC filter	Value of the filter to eliminate disturbance introduced by the electrical mains. The value must correspond to the mains frequency (e.g. 60 Hz in the USA and 50 Hz in Europe).	Off / 50 Hz / 60 Hz
HD+ sampling freq.	Sampling frequency of the HD+ acquisition device (the higher the frequency, the higher the bandwidth of the signal).	500 / 1000 Hz (1000 Hz *Windows* version only)
Antialias filter	Filter which smooths out the signal	On/Off
Rhythm lead	Selects the continuous trace to be displayed in the real time display window	12 leads: v1 / v2 / v3 / v4 / v5 / v6 / I / II / III / aVR / aVL / aVF /-aVR (inverted aVR) 15 leads: v1 / v2 / v3 / v4 / v5 / v6 /E1 / E2 / E3 / I / II / III / aVR / aVL / aVF /-aVR (inverted aVR)
QRS sound	Turns the heartbeat sound on / off. If on, the real time display window beeps at every heartbeat.	On/Off
Unit of measurement of patient height	Units of height (cm, in, mm)	cm/in/mm
Unit of measurement of patient weight	Units of weight (kg or lb)	kg/lb
Display race	Visibility of the race field in the patient's data and in print.	Yes/No
Reservation number	Display/modify the reservation number field	Concealed / Print only /

Field	Function	Possible values
		Modifiable

8.2.2. ECG

Field	Function	Possible values
Muscle filter	Filter to reduce muscle noise applied to the tracings.	Off / 25Hz / 40Hz / 150 Hz (150 Hz ***Windows*** version only)
Automatic interpretation	Selects whether and what type of automatic interpretation to use. If off, the interpretation is not included in the examination and in the print. If "short" the short strings are used, if "extended" the long strings are used. Only available if touchECG has the Interpretation option.	Off / Short / Extended
Show examination status	Selects whether to include "Report confirmed by..." or "Report not confirmed" in the examination	On / Off
QTcB	Activates the QT corrected calculation of Bazett	On/Off
QTcF	Activates the QT corrected calculation of Fredericia	On/Off
Auto Stat	Automatically sets the urgent flag for all examinations	On/Off
Automatic bookmark (min)	Sets automatic bookmarking during acquisition. Indicates the interval between bookmarks.	Numerical field
Department name	Name of the department (included in the printout and the examination)	Alphanumeric text field
Disable the lead fail screen	Allows turning off the display of the square waves in case of lead fail, so that you can see the real signal. Useful for patients with high impedance or with under-performing electrodes.	On/Off

8.2.3. Manual (*Windows* version only)

Field	Function	Possible values
Lead sequence	Order in which tracings are displayed and printed.	Standard / Cabrera
Trace format	Tracings display format.	12 leads: 12x1, 6+6, 3x1

Field	Function	Possible values
		15 leads: 15x1, 12x1, 6+6, 3x1
Speed mm/s	Trace speed in display and manual printout.	50/25/10/ 5 mm/sec
Amplitude mm/mV	Trace amplitude in display and manual printout.	20/10/5/2.5 mV/mm
Grid	Selects the type of grid to use in printouts.	Blank / Partial / Complete
Grid colour	Selects the colour of the grid to use in printouts.	Colour / Black and white
Output	Allows to choose the type of output during the manual printout	Printer/PDF/Both (default Printer)

8.2.4. Auto

Field	Function	Possible values
Printing format	Automatic mode trace printout format	12 leads: 12x1 / 6x2 / 3x4 / 3x4+1 / 3x4+3 / 12 complex 15 leads: 15x1 / 12x1 / 6x2 / 3x4 / 3x4+1 / 3x4+3 / 12 complex
Rhythm lead 1	1st rhythm lead (3x4+1 e 3x4+3)	12 leads: From I to V6 (default II) 15 leads: From I to E3 (default II)
Rhythm lead 2	2nd rhythm lead (3x4+3)	12 leads: From I to V6 (default V1) 15 leads: From I to E3 (default V1)
Rhythm lead 3	3rd rhythm lead (3x4+3)	12 leads: From I to V6 (default V5) 15 leads: From I to E3 (default V5)
Speed mm/s	Auto printout traces speed.	25 / 50 mm/s
Automatic print	Automatic printing of examination at the end of acquisition in automatic mode.	On/Off
Extra copies *Windows version only*	Number of printed copies. If 0, only the original copy is printed, if 1 the original + 1 copy are printed, if 2 the original + 2 copies are printed, and so on.	0 - 5 (default 0)
Add Median Page	To include the Median page and the vector ECG in the exported PDF document.	On / Off
Save exam in filtered mode	Save the exams in filtered mode, with this mode it will no longer be possible to change the muscle filter once saved	On / Off

Field	Function	Possible values
Median page in PDF	Includes median page when exporting and saving PDF	On / Off (default Off)
Extra Lead	Allows you to select the type of extra leads available only with HD + 15 and CLICK ECG-HD15	Rear / Right / Pediatric / Custom
Extra Lead # 1	In case of Custom type selection it is possible to select the lead. Selection box of the available Leads	V7 / V8 / V9 / V3R / V4R / V5R
Extra Lead # 2	In case of Custom type selection, it is possible to select the lead. Selection box of the available Lead	V7 / V8 / V9 / V3R / V4R / V5R
Extra Lead # 3	In case of Custom type selection, it is possible to select the lead. Selection box of the available Leads	V7 / V8 / V9 / V3R / V4R / V5R

8.2.5. Connectivity

Field	Function	Possible values
Device #	Device ID	Alphanumeric text field
Department ID	Department ID It lets you enter one Ward/Department ID only or several IDs, each separated by the characters ',', '\', '\n', ' ', '|' or ';'.	Alphanumeric text field
Boot Screen	It lets you start TouchECG with a screen for manual Ward ID entry (see para.5.2.1). It is disabled by default.	On/Off
Automatic transmission	Allows to automatically transmit an exam when it’s saved. If enabled, the automatic save function is enabled as well.	On/Off
Delete test after sending	This allows the test to be deleted after having been sent successfully.	On/Off
Receiving protocol	Sets the work list receive protocol.	Blank / GDT / Cardioline / Text file / XML
Transmission protocol	Sets the examination transmit protocol.	Blank / GDT / Cardioline / XML
Protocol configuration	Sets which protocol to configure. Which fields display depends on this parameter.	Blank / GDT / Cardioline / Text file / XML
Note: *DICOM connectivity is achieved by using “Cardioline” protocol and configuring it as described in par. 7.6.3.*		
Protocol configuration – GDT (*Windows* version only)		
Work list folder	Path of the folder containing the work lists created by the GDT system (selectable with the mouse using the "Browse" button).	Alphanumeric text field

Field	Function	Possible values
Examination folder	Path of the folder in which touchECG saves the GDT format examinations, for uploading to the GDT system (selectable with the mouse using the "Browse" button).	Alphanumeric text field
Attach PDF examination	If enabled, allows you to attach a PDF examination report to the GDT examination file	On/Off
Protocol configuration – Cardioline		
Connection URL	Standard connectivity: Address of the server running Cardioline ECGWebApp to which you wish to connect DICOM connectivity: IP address	Alphanumeric text field
Username	Standard connectivity only: username with which touchECG automatically accesses Cardioline ECGWebApp	Alphanumeric text field
Password	Standard connectivity only: password for the username with which touchECG automatically accesses Cardioline ECGWebApp	Alphanumeric text field
Protocol configuration – Text file (*Windows* version only)		
Work list folder	Folder containing the text format work list (selectable with the mouse using the "Browse" button).	Alphanumeric text field
Format	Format of the work list (only "Standard" is currently available)	Standard
Protocol configuration – XML (*Windows* version only)		
Test folder	Path of the folder in which touchECG saves the test file in XML format (selectable with the mouse using the "Browse" button).	Alphanumeric text field

8.2.6. Other

Field	Function	Possible values
SCP auto export	Sets automatic export to the specified folder of the SCP format examination file after acquisition.	On/Off
SCP folder	Path of the folder to which the SCP files are exported (selectable with the mouse using the "Browse" button).	Alphanumeric text field
PDF auto export	Sets automatic export to the specified folder of	On/Off

Field	Function	Possible values
	the PDF format examination file after acquisition.	
PDF folder	Path of the folder to which the SCP files are exported (selectable with the mouse using the "Browse" button)..	Alphanumeric text field
Filename format	Sets the format of exported file names, with the following options: {ID}_{PATIENTID}_{FIRSTNAME}_{LASTNAME}_{GENDER}_{BIRTHDATE}_{AGE}_{TIMEOFACQUISITION}_{ACCESSIONNUMBER}_{CART}_{CONFIRMED}	Alphanumeric text field
Database path *Windows version only*	Path of folder in which the program's database is created and saved (selectable with the mouse using the "Browse" button).	Alphanumeric text field
Virtual keyboard	Turns the virtual keyboard on/off.	On/Off
Send statistics / errors	Enables you to send the log file automatically to the Cardioline server for assistance and support.	On/Off
Launch application *Windows version only*	If set, configures the system to launch an external application when the respective button is pressed in the real time display window. The value is the path of the application in question.	Alphanumeric text field
Automatic DPI	Enables the program to adapt the graphics automatically according to the DPI of the monitor being used. If it is disabled it uses standard as default.	On / Off
Quick user interface	For quick viewing of the display window in real time, in which only a few buttons are shown (see par. 4.5).	On / Off
Kiosk mode *Windows version only*	If enabled, it allows you to launch the application in “Kiosk” mode, i.e. full screen with full PC control (no other application can be used at the same time). *Available only on **Windows 10** operating system.*	On / Off
E-mail address	This allows the email address which the tests are to be emailed to, to be changed. By default, the address is: noreply@yourdomain.com.	Alphanumeric text field

8.2.7. License

Field	Function	Possible values
HD+ license	Checks whether the connected HD+ unit is licensed for touchECG.	Read only
Type of interpretation	Indicates the active interpretation type: none (no automatic interpretation available) or full analysis (Glasgow)	Read only
Connectivity	Indicates the active connectivity option: standard (standard internet connectivity) or DICOM (compatible with the DICOM protocol)	Read only
Privacy	Indicates the active option for the privacy mode, which enables the features described in paragraph 9.	Read-only
Cardioline Dongle	Indicates whether the Cardioline HD+ Dongle is connected to touchECG.	On/Off
Activation code	Allows you to enter a code to update the active options (par.10)	Alphanumeric text field

8.2.8. Safety

Field	Function	Possible values
Safety	Allows you to enable the protection of the settings. If “Security” = On when accessing the settings window, the PIN must be entered	On / Off
PIN	Security code	Numerical text field (4 digits)
Verification PIN	Confirmation of the security code	Numerical text field (4 digits)

8.3. Protection of the Settings

You can set a security code (PIN) that allows access to the settings section to be protected. In this way you can prevent the settings from being changed accidentally or by unauthorised personnel.
Proceed as follows to enable the protection of the settings:

- set the "Security" field to "On"
- enter the digits of the security code in the "PIN" field
  If the value entered in the "verify PIN" field differs from the one entered in the "PIN" field, an error message appears in the lower bar of the application and the settings cannot be saved
- enter it again to confirm the value of the security code in the "verify PIN" field

The security code must be numerical and a maximum of 4 digits.

Click on **Save** to save this setting.

Subsequent log-ins to the Settings section will require the PIN to be entered, as shown in the figure, which must be entered in the text field shown.

Buttons

- 

OK — Confirm the PIN and if correct, it allows you to access the Settings.
If the PIN is not correct, an error message is shown in the lower bar of the application
-  Back — Closes the window without opening the Settings.

8.4. Virtual Keyboard Management (*Windows* version only)

Recommended parameters are provided below to manage the virtual keyboard correctly to use the touchECG.

VERSION	DEVICE	SETTINGS
Windows 10 Anniversary	Touchscreen system with no physical keyboard	▪ Disable the "Virtual keyboard" setting from touchECG

VERSION	DEVICE	SETTINGS
		▪ Enable the "Tablet mode" or "Show virtual keyboard" when tablet mode is not enabled from Windows (refer to the Microsoft Windows guide)
Windows 10	Touchscreen tablet with no physical keyboard	You can choose whether to control activation of the virtual keyboard from the touchECG or from the operating system. In the latter case, follow the instructions such as in the Windows Anniversary Edition. If you want to control the opening of the virtual keyboard from the touchECG, make the following settings: ▪ Ensure the "tablet mode" is disabled from the Windows settings. ▪ Always from the Windows settings, disable "Show virtual keyboard when tablet mode is not enabled" ▪ Enable "Virtual Keyboard" from the touchECG settings

9. SET THE DEVICE IN ACCORDANCE WITH GDPR (General Data Protection Regulation) – *Windows only*

9.1. General information

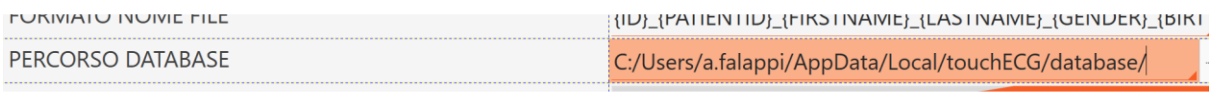
touchECG, Windows version from v. 3.40 onwards – if fitted with the Privacy option, it provides a series of features (described in the table below) that can be used by the Data Controller in order to comply with the minimum requirements of EU Regulation 2016/679, known as GDPR (General Data Protection Regulation).

GDPR requirement	Solution
Access control	Through the use of username and password at Operating System level and installing an instance (each one with its own database to which only the corresponding user can access) of touchECG for each user.
Data at rest protection	By the system administrator, activating the encryption features of the operating system.
Audit trail	Through the log of the Windows operating system, which traces the operations performed on the system. Moreover, touchECG allows you to reduce the number of tests saved in the local memory to limit data exposure.
Patient data removal (right to be forgotten)	It is possible to delete the examinations from the archive and to activate the automatic cancellation of the examinations after the transmission (where the use scenario foresees it).

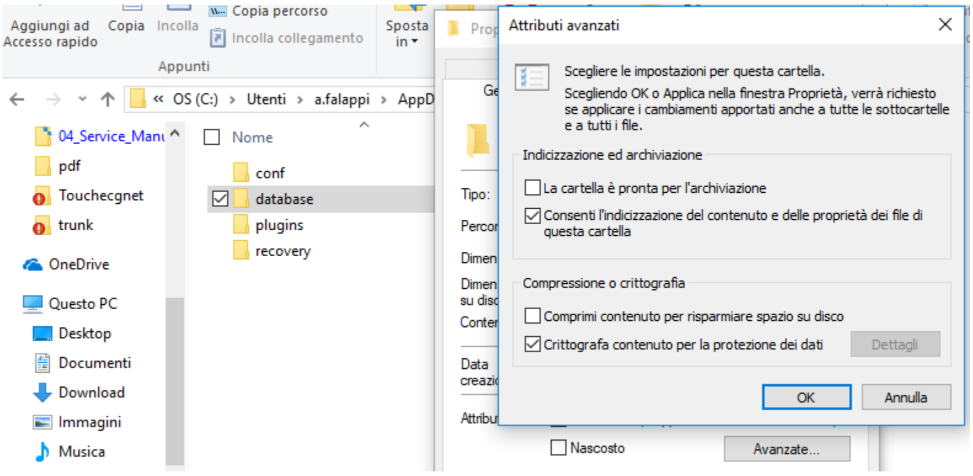
9.2. Encrypt the folder containing the database

Proceed as follows to encrypt the folder containing the touchECG database:

- Locate the database path from the Settings window > OTHER > "Database path";



- Using the Windows File Explorer tool, open the folder that contains the *database* folder, for example *C:\Users\a.falappi\AppData\Local\touchECG*;
- Select the *database* folder, right-click and select "Properties";
- Click "Advanced" and check the selection "Encrypt content.."



9.3. Enable the audit trail

The audit trail feature is automatically enabled if the software is normally installed from the CD through the Standard setup, whereas for the ClickOnce version, it must be enabled as follows:

- Download it from the website: http://update.cardioline.com/touchECG/EnableEventViewer.zip
- Extract the file into a local folder and run the “EnableEventViewer.cmd” command with administrative privileges;
- Check the touchECG option “Enable audit trail” in event viewer under “windows\application logs”.

9.4. Enable the “Delete test after sending” setting.

To enable the “Delete test after sending” setting, see par. 8.2.5.

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10. UPDATING THE SOFTWARE AND OPTIONS

10.1. Updating the software

The software is only updated if it has been installed with the Microsoft ClickOnce technology (for Windows) or from the website http://update.cardioline.com/touchECG or for Android through the Google Play Store. Aside from the software, the update process also includes the latest version of this user manual located in the installation directory.

10.1.1. Updating on Windows

For Windows systems, updates will be made available via the application setup file.

10.1.2. Updating on Android

For Android systems, updates will be made available via the installation APK file provided through an appropriate media.

10.2. Updating enabled options

10.2.1. General information

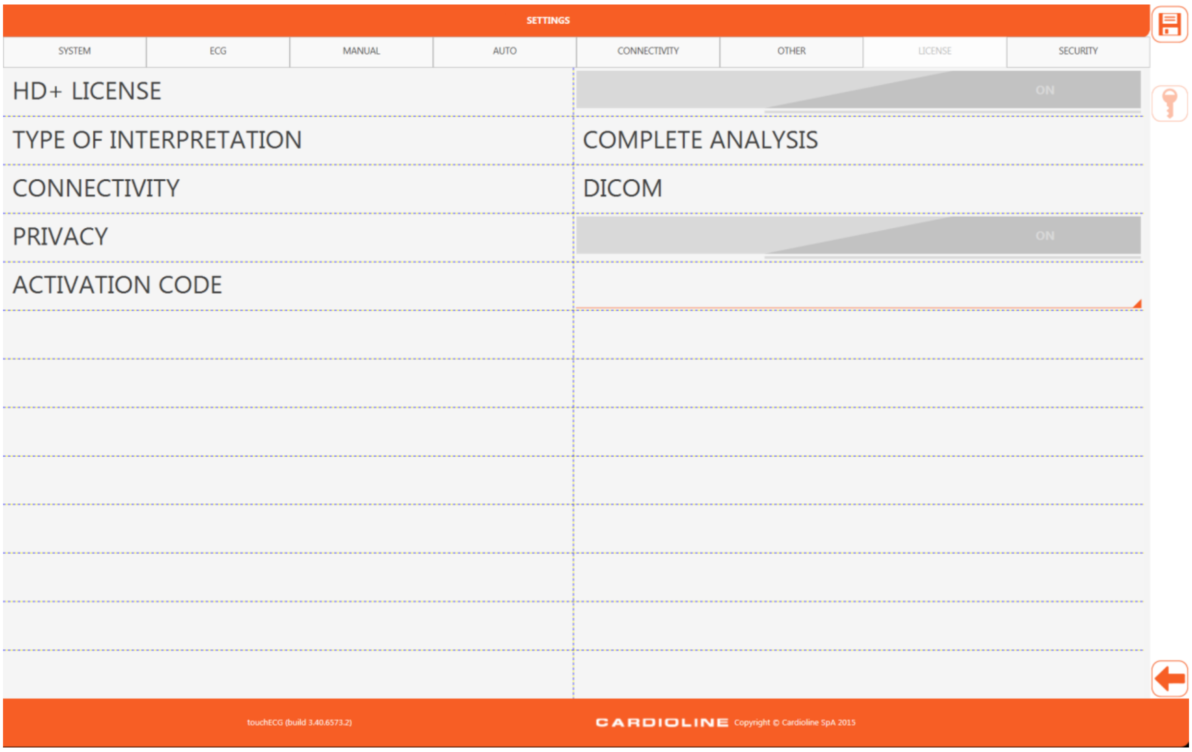
The options available on touchECG are controlled by the HD+ unit connected to it. Indeed, depending on the options enabled on the HD+, the corresponding touchECG options are enabled or not. It follows that depending on the acquisition unit in use and its enabled options, the touchECG may have different enable functionality.

In particular, the following options are available:

- HD+ license: enables the unit to connect to and communicate with touchECG
- Interpretation: enables the Glasgow automatic interpretation algorithm
- DICOM connectivity: enables connection to external systems using the DICOM protocol
- Privacy: this is used to enable the settings to guarantee compliance with GDPR (General Data Protection Regulation).

The options enabled on the HD+ unit connected to touchECG are summarised in the LICENZA (LICENSE) tab in Settings (par. 8.2).

__Note__: since the available functions depend on the HD+ unit you are using, these functions will be reflected in the acquired examinations.
For example, an examination acquired by an HD+ without the interpretation option enabled will not have an automatic interpretation, while an examination acquired to the same touchECG with an HD+ unit with the interpretation option enabled will have an automatic interpretation.
In the same way, an examination acquired by an HD+ unit without the DICOM connectivity option cannot be transmitted to a PACS system, while an examination acquired to the same touchECG but with an HD+ unit with its DICOM connectivity option enabled can be transmitted to a PACS system.



Settings window – LICENSE tab

10.2.2. Entering the activation code

To update the available options, and activate new ones, you must enable them on the HD+ unit.

To do so, request the activation code for the options in question from Cardioline SpA.
Once you have the code, connect the HD+ unit you wish to update to the computer and select it in the Settings window, as described in par. **Errore. L'origine riferimento non è stata trovata.**.
You can now open the LICENZA (LICENSE) tab and enter the activation code in the respective field. If the code is valid, the **Update license** button enables; click on it to start the update process.


***Update license** button*

When the update completes, the new enabled options are listed in the LICENZA (LICENSE) tab.

11. MAINTENANCE AND TROUBLESHOOTING

11.1. General information

touchECG is software and as such is maintenance-free.

For the maintenance of the HD+ acquisition unit, refer to its user manual.
In the same way, for the maintenance of the support device, refer to the user manual supplied with the computer.

If necessary, contact Cardioline SpA or an authorised distributor for assistance or to report malfunctions and anomalies.

***Note**: the computer should not be serviced in the patient area; make sure that you have taken all necessary precautions to eliminate electrical hazards (see par.* 2*).*

11.2. Operation check

You can check the operation of touchECG and its connection to the HD+ unit with an ECG simulator, to acquire a standard 12 lead ECG of known amplitude, so that you can check that the signal displayed by touchECG is conforming to the generated signal.

11.3. Bluetooth

No issues related to *"pairing"* the HD+ Cardioline device with the touchECG system have been found on tested systems.

Please also note that the correct start-up procedure of the HD+ system and touchECG is:

- first switch on the HD+ unit
- then start touchECG

This prevents having to place the HD+ device in error (fast flashing blue). If this problem arises it is advisable to wait for the HD+ to switch off automatically or remove the batteries and turn it back on.
If the Bluetooth signal is "lost", the user is warned by a message that appears.

If you repeatedly experience any Bluetooth related problems it is advisable to proceed as follows:

- Move the HD+ closer to the touchECG;
- Disable WiFi on the tablet/computer being used;

- Set the sampling rate to 500hz;
- Replace the BT antenna or, if internal, use an external Bluetooth key verified by Cardioline;
- Ensure both the HD+ device and the tablet have fully charged batteries, otherwise replace the batteries or charge them; both devices should always have the maximum available charge;
- Disable automatic updates on *Windows* systems
- Disable or remove antivirus and firewall client programs.

11.4. Troubleshooting table

Problem	Cause	Solution
touchECG does not connect to the HD+	Bluetooth signal weak or with interference	Approach the HD+ unit to touchECG
touchECG does not connect to the HD+	The HD+ is switched off	Turn it on
touchECG does not connect to the HD+	HD+ received an incorrect Bluetooth connection or the HD+ acquisition device flashes unexpectedly	Turn the HD+ unit off and back on again
When REALTIME launches, the HD+ unit turns off	The HD+ unit's battery is exhausted	Replace the battery

11.5. Messages and solutions table

Message	Cause	Solution
General errors		
Database full, delete some tests or the ones that have already sent will be automatically eliminated.	Examinations archive full	Delete some or all examinations from the Examinations Archive window to make space for new ones. If you are unable to delete the examinations, touchECG will delete them automatically, starting with previously transmitted ones
Database almost full, delete some tests.	Examinations archive almost full	Delete some or all examinations from the Examinations Archive window to make space for new ones and avoid automatic deletion.
The old transmitted examinations have been deleted	Automatic deletion of transmitted examinations due to the	Delete some or all examinations from the Examinations Archive window to make space for new ones and avoid automatic

Message	Cause	Solution
	Examinations Archive being full.	deletion.
Check the electrodes:	Bad contact of the skin with the electrodes	Check that the electrodes are properly connected to the patient.
HD+ SEARCH	The system is searching for the acquisition unit.	Information message. No problem associated with this message. Wait for the system to connect to the HD+ unit.
HD+ IS NOT LICENSED FOR TOUCHECG 3	The HD+ has been identified but it is not licensed for connection to touchECG.	Use another acquisition unit or contact Cardioline SpA for a license (par. 10)
HD+ NOT FOUND	No connection could be made with the HD+ unit.	The HD+ unit has not been found and no connection was made. Check that HD+ is switched on and near the computer.
HD+ BATTERY LOW, PLEASE CHANGE	HD+ battery low.	Replace the battery.
START REALTIME	Start of real time acquisition.	Information message. No problem associated with this message.
WAITING FOR AUTOMATIC ACQUISITION 10 SECONDS:	10s of signal not available for the examination	Information message. No problem associated with this message. Wait for the system to acquire 10s of signal.
WAITING FOR ACQUISITION OF AT LEAST 10 SECONDS:	10s of signal not available for printout	Information message. No problem associated with this message. Wait for the system to acquire 10s of signal.
Error settings NOT saved	The settings have not been saved	Check that the entries are valid and in the correct format
Database found!	touchECG has opened the application database	Information message. No problem associated with this message.
DATABASE UPDATED SUCCESSFULLY	touchECG has opened the application database	Information message. No problem associated with this message.
Database error check writing permits	The touchECG database is write protected	The database or folder does not have write permissions; try entering them.
Failed loading, error in archive	The examination in the database is corrupted	If the error persists, delete the examination from the archive
SYSTEM INITIALISATION	The touchECG system is booting up	Information message. No problem associated with this message.
Failed email, no email programme installed	Error in crating the email due to the fact that no email application has been configured on the computer	Configure an email program and try again.
PRINT ERROR, PLEASE CHECK PRINTER	The printer is not ready	Check that the printer is online and ready

Message	Cause	Solution
PRINTER NOT FOUND	No printer is configured in the system	Add a printer to the system
Not saved, no more space on disc or insufficient permits	The touchECG database has an error	Contact Cardioline support or delete the database (all saved examinations will be lost); touchECG will create a new one.
PDF creation error	File save error	Check the save path and permissions
Error on SCP file creation	File save error	Check the save path and permissions
FOLDER NOT FOUND, PLEASE CHECK PLUGIN SETTINGS AGAIN	Plugin configuration error	Check the settings of the plugins enabled in the touchECG settings
FILE NOT FOUND, PLEASE CHECK PLUGIN SETTINGS AGAIN	Plugin configuration error	Check the settings of the plugins enabled in the touchECG settings
PLUGIN CONNECTION NOT CONFIGURED	Plugin configuration error	Check the settings of the plugins enabled in the touchECG settings
GENERIC ERROR	Application error	Contact Cardioline support
NOTHING TO SEND	No examinations available to send	Acquire an examination and save it to the application database
PATIENT ACQUIRED	Working List acquired	Information message. No problem associated with this message.
CONNECTION IN PROGRESS...PLEASE WAIT	The system is connecting to an internal or external system	Information message. No problem associated with this message.
RECEIVED TESTS/UPDATED TESTS COUNT:	Examinations have been sent to an internal or external system	Information message. No problem associated with this message.
SENT TEST	Examination sent successfully.	Information message. No problem associated with this message.
SENT TEST COUNT:	Number of examinations sent	Information message. No problem associated with this message.
FILE NOT COMPATIBLE	The GDT configuration file is not compatible	Check that the file matches the specifications of para. 7.6.1.
PRINTING IN PROGRESS	The print job of an ECG test has been launched	Information message. No problem associated with this message. Wait for the printer to complete printing
WEAK BLUETOOTH SIGNAL	Communication is problematic with HD+	Error message. Move the HD+ device toward the touchECG. For the ***Windows version***, in case of continuing lost packets and if the sampling frequency is set at 1000 Hz, change the sampling frequency to 500hz. In any case check the instructions of par. 2.1.
NOT COMPATIBLE RADIO REQUIRES AT LEAST BLUETOOTH>	Bluetooth radio of host system not compatible	Change bluetooth antenna in host system having Bluetooth requirement> = 4.2

Message	Cause	Solution
= 4.2	with HD + 12/15 and CLICKECG-HD12 / 15	
RESOLUTION AVAILABLE ONLY FROM BLUETOOTH 5	To operate with HD + 12/15 and CLICKECG-HD12 / 15 high resolution it is necessary to have a Bluetooth antenna with 2MPhy option in the host system	Change bluetooth antenna in the host system with 2MPHY requirement

12. TECHNICAL SPECIFICATIONS

ECG acquisition (HD+ unit)	
ECG leads	12 or 15-leads (I, II, III, aVR-L-F, V1-6, E1-3)
Patient cable	10 or 13 replaceable wire patient lead
CMRR	115dB
DC input impedance	100MΩ
A/D converter	24 bit, 32000 samples/second/channel
Front-end sampling frequency	32000 samples/second/channel
Sampling frequency for signal analysis	Windows: 1000 samples/second/channel Android: 500 samples/second/channel
A/D conversion	20 bit
Resolution	<1 µV/LSB
Dynamic range	+/- 400 mV
Bandwidth	Windows: Performances equivalent to 0.05-300 Hz Android: Performances equivalent to 0.05-150 Hz
Pacemaker detection	Hardware detection coupled with digital convolution filter
De fibrillation protection	AAMI/IEC standard
Front-end performance	ANSI/AAMI IEC 60601-2-25:2011
Data transfer	Bluetooth 2.1+ EDR with "secure pairing" for HD+ Bluetooth >= 4.2 for HD+ 12/15 at 500 Hz Bluetooth >= 4.2 with 2MPHY for HD+ 12/15 at 1000 Hz
Processing	
Operating system	Windows / Android
Pacemaker detection	Hardware recognition (HD+ acquisition unit)
Lead-fail detection	Independent for all leads
Cardiac frequency range	30 - 300 bpm
Sampling frequency	Windows: 1000 samples/second/channel Android: 500 samples/second/channel
Base line stabilisation	Fully digital diagnostic high pass filter
AC filter	Adaptive 50/60 Hz digital filter
Muscle filters	Linear phase digital diagnostic high-pass filter (according to 60601-2-25 2nd ed.) 50/60 Hz AC interference adaptive digital filter Digital low pass filter (for printout and display):

	Windows: 25/40/150 Hz **Android**: 25/40 Hz
ECG acquisition mode	Automatic (12/15 leads), Manual (3/6/12/15 leads), Review (12/15 leads)
Lead configuration	Standard, Cabrera
ECG measurements	All leads, medians, corrected HR Average RR PR Interval QRS Duration QT and QTc (Hodges formula) intervals QTc Bazett interval QTc Fridericia interval max R[V5];[V6] and S[V1] Sokolow-Lyon Index P, R, T axis
ECG interpretation	Glasgow algorithm for adults, paediatric, STEMI (optional)
ECG interpretation parameters	Race, sex, age, drugs
Memory	Internal archive up to 1000 ECGs
Processing Options	
Interpretation	Glasgow algorithm for adults, paediatric, STEMI
Connectivity	DICOM
Exported formats	
SCP-PDF-XML-GDT	Standard format
DICOM	Included in DICOM connectivity option
HL7	Optional
Connectivity	
USB-LAN-WiFi	Dependent on support device (computer)
Printing	
Resolution	Variable in relation to printer
Paper type	Variable in relation to printer
Sensitivity/gain	5, 10, 20 mV/mm
Automatic print speed	25, 50 mm/s
Automatic print	3, 3+1, 6, 12 channels; Standard or Cabrera;
Automatic print formats	12-lead: 12x1, 6x2, 3x4, 3x4+1, 3x4+3 15-lead: 15x1, 6x3+vector ecg, 3x5, 3x5+1, 3x5+3
Manual print speed	5, 10, 25, 50 mm/sec
Manual printing	3, 6, 12, 15 (only HD+ 15 and CLICKECG-HD 15) channels; Standard or

	Cabrera;
Manual print formats.......................	12x1, 6+6, 3x1, 6x3, 15x1 (only HD+ 15 and CLICKECG-HD 15)

12.1. Filter features

The device adopts different filtering techniques in order to process the signal and make it easier for the cardiologist to make the diagnosis.

The device implements a high-pass linear-phase filter, with a cutoff frequency of 0.67Hz, to remove the drift of the entirely digital baseline.

The filter effectively reduces the artefacts induced by the respiratory movement without introducing distortions on the reproduction of the ST segment, as specifically recommended in "cf. Paul Kligfield et al., Recommendations for the Standardization and Interpretation of the Electrocardiogram Part I, Circulation. 2007;115:1306-1324”:

> *To reduce artifactual distortion of the ST segment, the 1990 AHA document recommended that the low-frequency cutoff be 0.05 Hz for routine filters but that this requirement could be relaxed to 0.67 Hz or below for linear digital filters with zero phase distortion. The ANSI/AAMI recommendations of 1991, affirmed in 2001, endorsed these relaxed limits for low-frequency cutoff for standard 12-lead ECGs, subject to maximum allowable errors for individual determinants of overall input signal reproduction. These standards continue to be recommended*

The high-pass filter meets the requirements set by IEC 60601-2-25 2nd Ed. in terms of low-frequency impulse response:

> *“A 0,3 mV × s (3 mV for 100 ms) impulse input shall not produce a displacement greater than 0.1 mV outside the region of the impulse.”*

The acquisition (sampling and filtering) system of the device complies with AHA recommendations (Paul Kligfield et al., Circulation 2007) for both paediatric and adult ECGs. The high frequency response of the system is 150Hz or 300Hz, depending on the filter applied.

The device also offers the option of applying, only in display and printing mode, digital linear-phase filters with a cutoff frequency of 25Hz or 40Hz, which reduce the bandwidth of the printed signal in order to reduce the effects of high frequency noise ("muscular" noise). As a consequence of the application of these filters, the resulting signal no longer meets the minimum requirements on the high frequency response reported in the above recommendations.

The device can also be configured to selectively eliminate interference induced by the electricity network (50Hz or 60Hz, depending on the country where the device is installed).
The network filter complies with the requirements of IEC 60601-2-25 2nd Ed.

12.2. Harmonised standards applied

STANDARD	DESCRIPTION
EN ISO 15223-1	Medical devices - Symbols to be used with medical device labels, labelling and information to be supplied - Part 1: General requirements
EN 1041	Information supplied by the manufacturer of medical devices
EN ISO 13485	Medical devices - Quality management systems - Requirements for regulatory purposes
EN ISO 14971	Medical devices - Application of risk management to medical devices
EN 60601-2-25	Medical electrical equipment - Part 2-25: Particular requirements for the basic safety and essential performance of electrocardiographs. *Partly applied – Applied in conjunction with HD+*
IEC 60601-1-11	Medical electrical equipment -- Part 1-11: General requirements for basic safety and essential performance -- Collateral standard: Requirements for medical electrical equipment and medical electrical systems used in the home healthcare environment. *Partly applied – Applied in conjunction with HD+*
EN 62304	Medical device software - Software life cycle processes
EN 62366	Medical devices - Application of usability engineering to medical devices

12.3. Accessories

CODE	DESCRIPTION
81018027	HD+
81018228	HD+ 15
81018231	HD+ 12
81018328	HD+ 15 (blue)
81018331	HD+ 12 (blue)
81018428	CLICKECG-HD 15
81018431	CLICKECG-HD 12

13. WARRANTY

Cardioline SpA guarantees this device for a period of 24 months from the date of sale. The date of purchase shall be proven by a document, issued upon delivery, which shall be submitted in the case of any claim under the warranty.
The warranty covers restoring the device operation free of charge for malfunctions relating to installation or the software itself.
This warranty does not cover defects resulting from:

- tampering, third party negligence, including servicing or maintenance by unauthorised personnel;
- failure to comply with the user instructions, improper use or use of the equipment other than that for which it was manufactured;
- damage caused by fires, explosions or natural disasters;
- use of unauthorised software;
- other circumstances not attributable to manufacturing defects.

Cardioline Spa declines all liability for any damage which may be caused, directly or indirectly, to persons or property as a consequence of non-compliance with all the prescriptions specified in the manual, especially warnings regarding installation, safety, use and maintenance of the equipment, as well as non-operation of the equipment.
For restoration/repairs, contact Cardioline SpA or an authorised service centre. Labour is free of charge while the risks and costs of transport are borne by the customer.
The warranty expires after 24 months from the date of purchase; any subsequent assistance will be charged at normal labour rates.
Any derogation from the present warranty conditions shall be valid only if expressly approved by Cardioline SpA.

Head Office and Production
Via Linz, 151
38121 Trento
Italy
T. +39 0463 850125
F. +39 0463 850088

Sales Office:
Via F.lli Bronzetti, 8
20129 Milan, Italy
T. +39 02 94750470
F. +39 02 94750471

CARDIOLINE